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The (A)iceberg beneath tech debt: Recognized calm, rising spreads below the waterline

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In Summary

- **Equity captures the upside of the AI build-out, while credit absorbs the loss if it fails – yet creditors are compensated with under 1% p.a. on 5-year bonds and around 2% on 10-year.** The asymmetry has grown as total long-term debt at the eight dominant US tech companies rose by 86% in the 12 months to mid-2026. If AI investment pays off, equity holders benefit; if it does not, the added debt, increasingly compounded by off-balance-sheet commitments, leaves creditors exposed to a skewed distribution of outcomes that current spread levels may not reflect. This paper tests whether spreads price that skewed risk using three structural credit models – Merton, and KMV-inspired, and CreditGrades – with financial-filing and equity data for eight mega-caps.
- **On recognized debt, the market's calm is justified but we see the first drifts.** Our modelling agrees with market pricing and ratings on most names. As of mid-2026, recognized leverage across the panel is low, at 0.2% to 6.8%, implying credit risk broadly consistent with Aaa–A ratings. Three of the eight screen weaker than their formal rating due to high leverage (Meta) or equity volatility (Oracle, SpaceX).
- **The real risk lurks below the waterline: off-balance-sheet debt lifts the debt burden by nearly 150% on average, and pulls the model-implied credit quality down 1-2 notches.** The deterioration surfaces as quality migration before it surfaces in price, because distance-to-default is too far for spreads to react. The market has noted the direction, but not yet the size: CDS spreads more than doubled over the past year, while the average model-implied credit spread rose six-fold once off-balance-sheet debt is included. Off-balance-sheet debt keeps growing as new commitments are signed, and AI profit potential, even realized, offers little spread-tightening against it.
- **Ranked against the broad corporate credit universe, the panel is bifurcated:** model-implied credit quality spans from the 97th percentile (strongest) to the 2nd (weakest), with the average sitting near the 62nd — barely better than a median credit. In effect, the rank bought by rising market capitalizations and compressed volatility is being spent on debt growth, both recognized and hidden. The two legs are not symmetric – debt is contractual and accumulates on schedule, while equity relies on the future.
- **A severe market repricing – equity down 10%, volatility up 10pp, debt up 50% – would cost the panel two standard deviations of distance-to-default, translating into ~20bps of model spread.** Debt sensitivity is actually lower: recognizing leases – equivalent to 150% of debt – moves spreads by only ~10bps, with wide dispersion across names. Mid-2022 is the natural stress test: the same balance sheets, carrying far less debt, still showed vulnerability – volatility and time horizon matter more than rates or equity levels.
- **As tech allocation in credit portfolios grows, spreads do not fully compensate for skewed risk.** Default risk is still remote, but spread risk will increasingly surface. The liability leg is contractual and arrives on schedule, while the equity leg can reverse in a quarter. Tech's curve steepness already prices the long horizon as the key risk, and that may extend to the front-end. Tech has avoided the iceberg so far, but the navigation is getting trickier, and the metric to watch is spread vs. quality ranking within the corporate market, including by financial commitments and equity volatility.

From the broad market to the (A)Iceberg

What is the right price for carrying the tail risk of the AI build-out? We put that question to structural credit models that price default risk from the equity market and the balance sheet, rather than from credit markets themselves. By definition, equity keeps the upside if the AI build-out works, credit mainly absorbs the loss if it does not. The mega-cap technology complex already dominates the equity indices and is now ballooning in credit to finance that build-out. And investment-grade spreads offer only modest compensation of ~80bps, even after the tech-led widening of 2026.

The starting point is Merton (1974), and behind the mathematics sits a simple idea: a company owns assets and owes debt. If the assets are worth more than the debt when the bill comes due, shareholders keep the difference; if they are not, creditors take what is left. Owning the shares is therefore a bet on the assets clearing the debt – and that means the equity market is already saying something about default risk. Two things drive the answer: how much cushion sits between what a company is worth and what it owes, and how fast that cushion can move, which is what equity volatility measures. A thick cushion that barely moves is safe. A thin one that swings around is not. The model implicitly assumes the consensus AI narrative embedded in equity, with equity volatility taken as a proxy for the market's uncertainty. This approach may not fully appreciate the income from current AI-investment. But credit investors will barely benefit from it, and instead need to focus on downside against slim spread.

Merton's limitation is that it is too well-behaved at the top end. Give it a company with 2% leverage and it reports a default risk of essentially zero – hard to argue and work with. A KMV-inspired approach¹ – our own implementation of the published structural framework, not the proprietary Moody's Analytics KMV model – keeps that logic and fixes the last step: instead of reading default risk off a bell curve, it reads it off what has actually happened in the past to companies with similar cushions. It also assumes a that firm has to cover its short-term debt plus part of its long-term debt, rather than everything all at once. The output arrives in a language credit investors already use – an implied rating. That matters here, because when Merton has flattened to zero, the rating still moves. CreditGrades, a variant that allows for uncertainty about exactly where the default point sits, is run alongside as a sanity check.

We run both models on key technology companies – Microsoft, Apple, Alphabet, Amazon, Meta, Nvidia, and Oracle. SpaceX was also added in recent analysis, completing the eight-name panel we refer to throughout. The analysis covers five timestamps from mid-2022 – the onset of the AI theme – to mid-2026, using the market capitalization, equity volatility and risk-free rate prevailing at each date. Each name is priced under two debt barriers: reported balance-sheet financial debt only, versus balance-sheet debt plus off-balance sheet commitments – together, the (A)Iceberg.

On the recognized balance sheet, default risk pricing is a volatility story

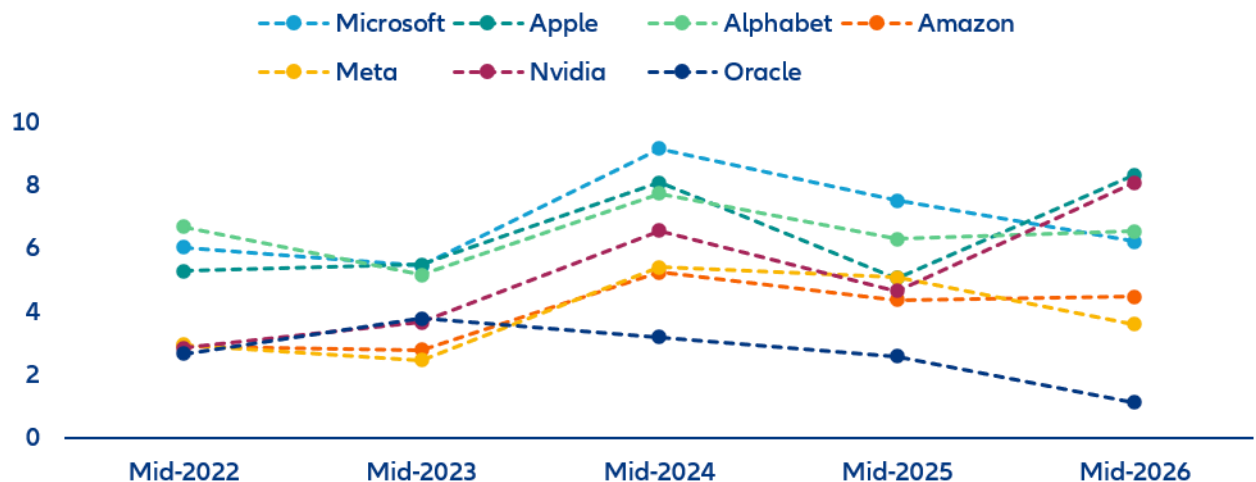
Five years of distance-to-default read as a volatility cycle, not a leverage cycle. Distance-to-default was weakest in mid-2022, when the inflation-and-policy shock had left market capitalizations 40–70% below today's levels and pushed single-name volatility into the 41–57% range. As the 2024 economic rally rebuilt equity cushions, it subsequently dipped in 2025, and split in 2026 (Figure 1). Meanwhile, balance sheets remained more static and the debt barrier quite stable from 2022 to 2025. Heavy issuance of new debt started only in late 2025. Oracle's path is the only one that can be explained through leverage: its distance-to-default has ground down substantially (from 2.4 to 0.8 standard deviations), as debt grew into an equity cushion that halved over 2026. Generally, below 1 standard deviation is naturally the danger zone (CCC-rating and below) while above 3 enters investment grade and 7 in AAA-space.

¹ Moody's KMV (Kealhofer, McQuown, Vasicek) operationalises the Merton model by mapping distance-to-default to default frequencies estimated from a proprietary default database, rather than through the Gaussian mapping, which understates default risk in practice. See Crosbie and Bohn (2003), Modeling Default Risk, Moody's KMV.

The 2026 differentiation is itself informative: model default risk improved sharply at Apple and Nvidia, while it deteriorated at Microsoft, Meta and Oracle, whose distance-to-default fell by 23–31%. Applying equity volatility has already divided the panel into the names it believes are winning the build-out and the names paying for it, with Oracle as the outlier – a point that debt markets also noted.

Figure 1: Distance to default, mid-2022 to mid-2026 – a volatility cycle for six names, a leverage story for one

Model-implied distance to default, standard deviations of asset value



Sources: Bloomberg, Allianz Research. Merton distance-to-default (standard deviations, 5-year horizon), recognized-debt barrier; higher is safer.

Mid-2026 shows the recognized numbers justifying the market's calm (Table 1), while SpaceX is repeating the volatility story. Recognized leverage – all short-term debt, plus 75% of long-term debt, against market-implied asset value – sits between 0.2% (Nvidia) and 6.8% (Amazon) for the six mega-caps; only Oracle, at roughly 25%, carries leverage that a structural model can even see. SpaceX makes the volatility point in its purest form: at just 1.3% leverage, its distance-to-default is short at 0.9. The model reads B against current ratings at Baa1, purely because equity volatility near 113% makes even a small barrier reachable.

Merton and CreditGrades spreads accordingly collapse for the AA/AAA quality names – nearly zero implied spread against market CDS of 31–71bps. Those numbers are not a data problem, they follow from what each side measures. Default risk at these balance sheets is so remote that the models' risk-neutral normal tail rounds it to zero, while the market spread is largely everything a default model does not price: liquidity, risk premium and the technicals of supply and demand. Model and market spread levels therefore cannot be compared directly – the information sits in the rankings and the changes.

The KMV-inspired overlay – an empirical mapping from distance-to-default to observed default frequencies, is used precisely to restore that information: Aaa-equivalent for Microsoft, Apple, Alphabet and Nvidia, A for Amazon, Baa for Meta and B for Oracle and SpaceX (Table 1). Set against the rating agencies, the mapping lands close for most of the panel: on five of eight companies, the model echoes both the agencies' and the market's calm. The outlier is Meta -Baa in the model against Aa3 at Moody's, roughly three notches below the agencies, because the model assumes 39% equity volatility and ignores the profit potential which is uncertain. Oracle's gap (B against Baa2), runs the same direction and the high leverage weighs on the result. However, the model better justifies market CDS price rather than the agencies: at 71bps Meta trades the widest of the six mega-caps excluding Oracle – wider than higher-rated Alphabet (Aa2, 55bps) and even BBB benchmark (60bps). The mismatch between market pricing and model implied default risk reflects the debt beyond the current balance sheet, namely announced future issuance and off-balance sheet debt, as well as future profit expectations.

Table 1: The recognized balance sheet in mid-2026 and model-implied rating

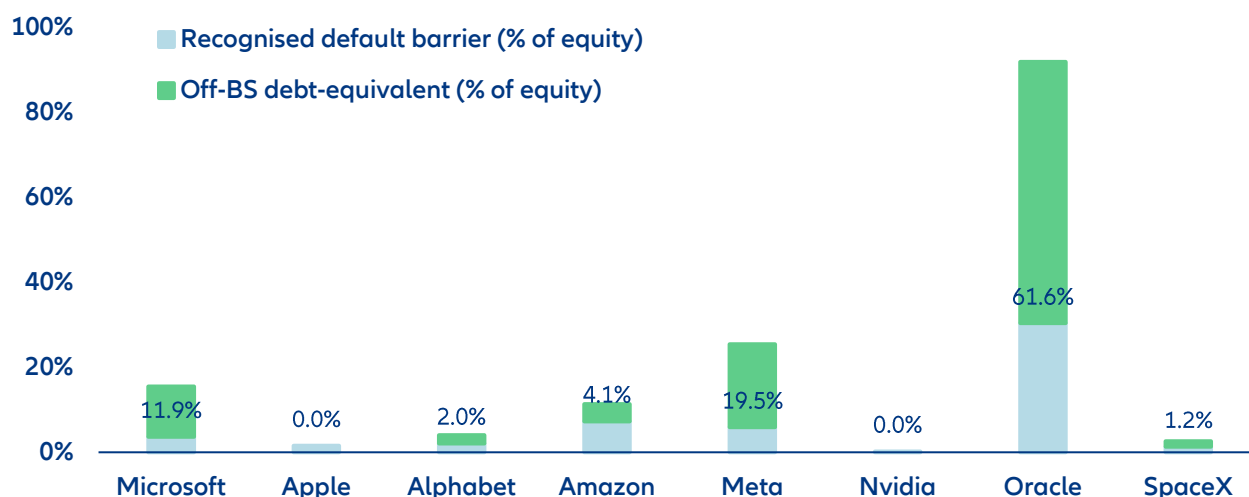
Company	Equity vol (1y, %)	Debt Barrier (USDbn)	Leverage	Distance to default	Model impl. rating	CDS impl. rating	Market 5y CDS at mid-2026 (bps)
Microsoft	27	101	3.6%	5.6	Aaa	Aa3	42
Apple	24	67	1.5%	7.9	Aaa	Aa2	31
Alphabet	30	90	2.0%	6.0	Aaa	A1	55
Amazon	31	186	6.8%	4.1	A	A1	55
Meta	39	85	5.7%	3.3	Baa	A3	71
Nvidia	36	10	0.2%	7.6	Aaa	A1	48
Oracle	63	128	24.9%	0.8	B	Baa3	171
SpaceX	113*	30	1.3%	0.9	B	Baa2	138

Sources: Bloomberg, Allianz Research. Barrier = ST debt + 0.75×LT debt; leverage = barrier / market-implied asset value; distance to default from the Merton solve at a 5-year horizon; model-implied rating from our own empirical default-frequency mapping. *Equity vol of Space X is annualized 10-days vol as there is no 1y trading history yet.

The (A)Iceberg: USD1.1trn of obligations the default barrier cannot see

The largest claims on future cash flow in this complex are hidden from the balance sheet. Data-center capacity contracted today but delivered in 2027–2031 stays disclosed-only until the lease commences: take-or-pay cloud, compute and power purchase agreements behave like debt services, but book as executory contracts; residual-value guarantees and vendor-financing structures sit further out still. According to Bloomberg Intelligence, Alphabet, Meta, Oracle, Microsoft, and Amazon collectively hold off-balance sheet commitments nearing \$2.6 trillion as of August 2026. That number is up from \$573 billion a year prior, driven primarily by AI infrastructure build-outs – including uncommenced data-center leases and long-term supply agreements.²

Figure 2: The iceberg in one chart – recognized barrier versus off-balance-sheet debt-equivalents, % of market equity, mid-2026

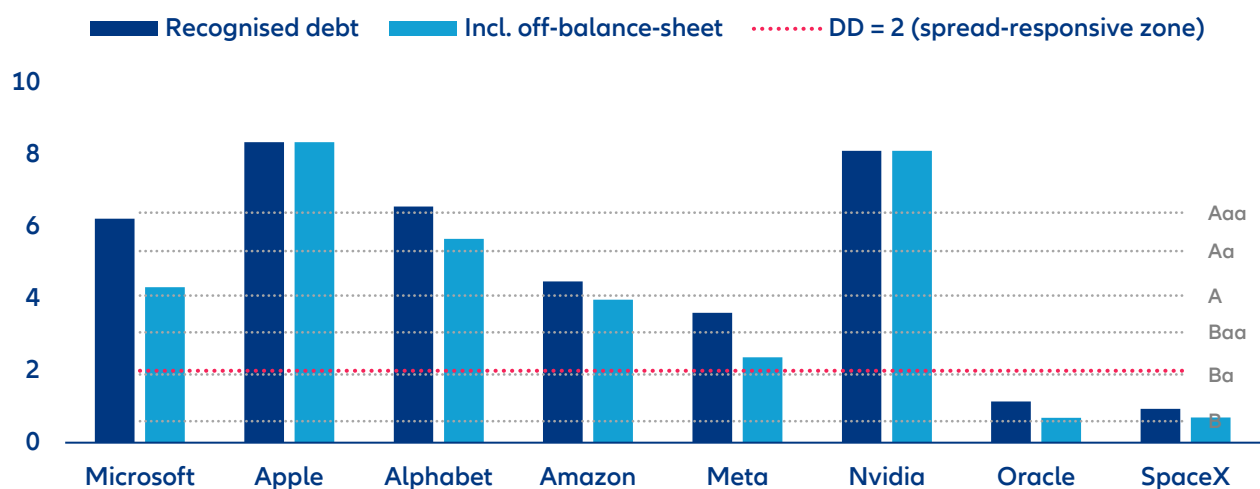


Sources: company filings (commitments notes), Bloomberg, Moody's, Allianz Research. Off-balance-sheet debt-equivalents are judgement-based estimates of not-yet-commenced leases; undiscounted, mid-2026 only.

² Bloomberg Intelligence Mandeep Singh, 08/13/26

The hidden debt moves model-implied credit quality by one to two notches while leaving most model spreads visually less changed. This is because the distance to default is still too far to reflect in model pricing. Adding the commitments lifts leverage to 16% and 25% respectively and multiplies Microsoft's and Meta's debt burden more than four times and (Figure 2). Therefore, as shown in Figure 3, distance-to-default would be cut by 1.9 and 1.2 standard deviations – equivalent to a 1-2 notch downgrade. Yet the model-implied spread response is close to nil everywhere except the two names whose cushions were thin to begin with: Meta moves from 0.2bps to 7.4bps (CreditGrades: 0.1 to 10.2bps) and Oracle from 192bps to 326bps (CreditGrades: 230 to 467bps). The model also translates the same compression into ratings language, full recognition of the off-balance sheet items moves model-implied quality by one to two notches overall (Microsoft Aaa to A, Alphabet Aaa to Aa, Meta Baa to Ba). Our measure of the stack is deliberately narrow: only uncommenced leases, the one component disclosed in the commitments notes and clearly definable as a debt-equivalent. Layering in the take-or-pay supply agreements, purchase obligations and guarantees would deepen every one of these migrations.

Figure 3: Distance to default, mid-2026 – recognized debt versus including the off-balance-sheet stack
Dotted line as model implied rating band floor

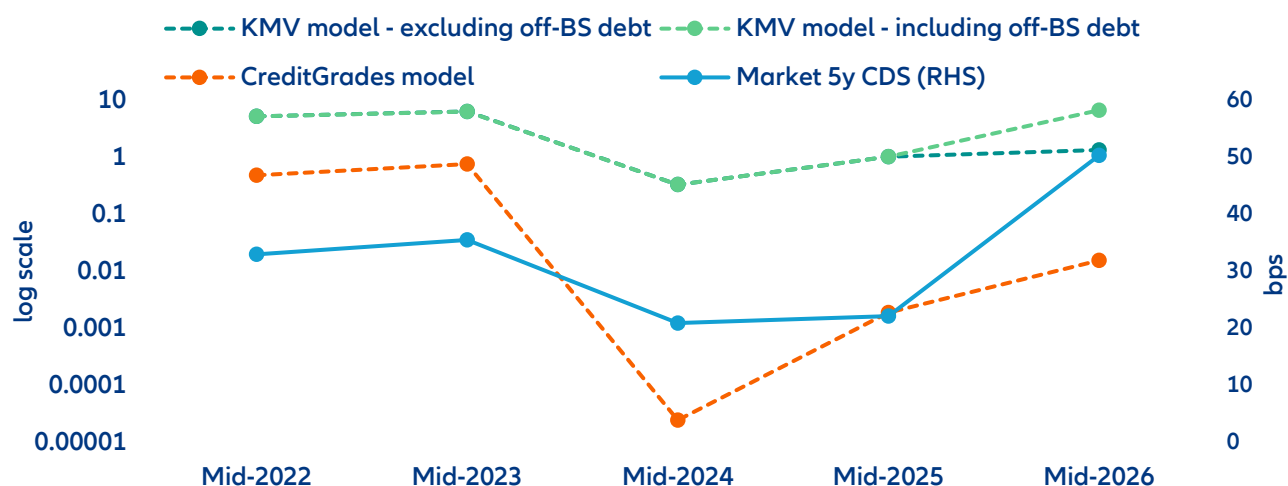


Sources: Bloomberg, company filings, Allianz Research. Below roughly $DD = 2$, structural model spreads become responsive; above it, barriers can multiply without spreads reacting.

The market is not blind to this – but it is not fully pricing it either. The 2026 widening is itself partial recognition: the two names carrying the largest commitment stacks relative to their recognized debt, Meta and Oracle, are also the two widest spreads in the group. Spreads have moved by tens of basis points year-to-date while, the inclusion of off-balance sheet items implies one to two notches of quality migration – a repricing consistent with part, not all, of the off-balance sheet stack. And the stack is not static: commitment schedules are still being signed, recognized debt is growing alongside them and every lease that commences converts a disclosed obligation into a recognized barrier.

Current spreads price part of the off-balance sheet stack, but pricing all of it would imply further widening, and the gap is still growing. Mid-2026 is the first time in the AI era that model and market widen together, and the models widen furthest where the hidden debt is counted. Model spread levels price expected default loss only, and sit far below traded spreads which reflect liquidity and other non-fundamentals drivers. Consequently, Figure 4 should be read for direction rather than level. From mid-2022 to mid-2025, model and market tightened together; into mid-2026 both widen. Once we exclude Space-X, due to its lack of history, and Oracle, due to its wider spread, the average market 5-year CDS of the remaining names more than doubled over the last year, from 22bps to roughly 50bps. The model average rises about 30% on recognized debt alone, and roughly six-fold once the off-balance-sheet stack is counted. This supports our argument that the market's widening only partially reflects inclusive deterioration. Given the continuing financing needs and dynamic nature of the commitments, we don't expect the widening pressure to fade away any time soon.

Figure 4: Model versus market, excluding Oracle, bps, log scale



Sources: Bloomberg, Allianz Research. Market line = average 5-year CDS; before 2026 only Microsoft, Apple and Amazon carried liquid quotes, and Alphabet, Meta and Nvidia enter the average in 2026. Model spreads capture expected default loss only and sit below market by construction – compare trends and the wedge, not levels.

Ranked against the broad issuer universe, the panel has bifurcated, stretching the whole range from 97th to 2nd percentile

Ranked against the broad issuer universe, rather than against each other, the panel has moved from clustered to bifurcated – and once the commitment stack is counted, four years of apparent improvement net out to almost nothing. On recognized debt, the average placement of the seven names rises from the 61st percentile in mid-2022 to the 73rd percentile in mid-2026, showing an improvement among the Investment Grade universe. Add the off-balance-sheet stack at full weight, however, and the average falls back to the 62nd (Table 2).

Table 2: Ranked against the broad issuer universe, the panel has bifurcated – percentile of model-implied distance-to-default, mid-2022 versus mid-2026

Company	Percentile mid-2022	Percentile mid-2026 ex off-BS	Direction on recognized debt	Percentile mid-2026 incl. off-BS	Net change (pp)	Direction incl. off-BS
Microsoft	95.0	94.1	Broadly stable	70.3	-24.7	Deteriorated
Apple	88.5	97.6	Improved	97.6	+9.1	Improved
Alphabet	96.3	95.4	Broadly stable	90.1	-6.2	Deteriorated
Amazon	39.0	73.8	Improved	62.0	+23.0	Improved
Meta	39.6	54.1	Improved	20.7	-19.0	Deteriorated
Nvidia	37.5	97.1	Improved	97.1	+59.7	Improved
Oracle	34.4	3.5	Deteriorated	1.8	-32.6	Deteriorated
Average	61.5	73.7	Improved	62.8	+1.3	Broadly stable

Sources: Bloomberg, company filings, Allianz Research. Percentile = placement of the model-implied physical distance-to-default in the broad issuer cross-section (323 issuers at mid-2022, 455 at mid-2026). Note: the comparator universes are built from recognized borrowings only, so the inclusive columns place an off-balance-sheet-adjusted distance-to-default into an unadjusted peer distribution – a conservative lower bound rather than a like-for-like rank. Direction reads the percentile change: improved or deteriorated for moves beyond ± 5 pp, broadly stable within it.

Although the aggregate level seems flat, the idiosyncratic movement is significant and the off-balance sheet debt amount is decisive. Three names improve on net, with or without the off balance sheet burden: Nvidia, Apple and Amazon all favorably ride favorable equity market effects with modest change in leverage. Nvidia gains most, from the 37th to the 97th percentile, and Apple rises from the 88th to the 98th. Neither carries a readily measurable commitment liability (Nvidia's buyer support via equity contributions obviously add circularity elsewhere and

financing support may rise in the future). Amazon keeps a net gain of 23pp despite handing back 12pp once its USD106bn of commitments are counted. Microsoft, Alphabet and Meta deteriorate only once the off-balance-sheet liability is counted, as their market-cap gains and volatility increases are not enough to offset liabilities growing several-fold. Microsoft is the clearest illustration: its placement is broadly stable over four years (95th to 94th) considering balance sheet debt, but if off-balance sheet liability is added, model-implied credit would deteriorate by one notch from Aa to A, as does the percentile rank. Meta runs the mechanism at higher intensity: +14 percentile-point change on recognized debt, -33 percentile-point from the stack, a huge net -19 percentile-points for eventual placement in the bottom quartile of the universe at the 20th percentile. Oracle is the only name dragged by everything at once – a fallen market capitalization, equity volatility, growing recognized debt and a stack equivalent to 60% of its equity – sliding from the 34th percentile to the 2nd at the bottom. All of above is not precise for each company but meant to offer insight into the dynamics for the overall technology panel and big-tech sector.

If benign markets turn: mid-2022 is the stress test built into the sample

Table 3 shows that horizon and equity volatility dominate the model results. Altogether, a stress scenario would cost the panel some two standard deviations of distance-to-default – equivalent to one to two rating letters or around 20bps of spread widening. Horizon matters the most (± 1 year is worth 0.42–0.58 standard deviations), because the model measures cumulative default risk within the horizon. The high horizon sensitivity also helps explain why the AI-related bonds curve is steeper, i.e. long-dated AI bonds are traded at a higher premium compared to the market curve. Among the market-facing parameters equity volatility dominates – 1pp moves distance-to-default by 0.20 standard deviations on average, against 0.11–0.13 for a 10% move in market capitalization or in debt. Implicitly, the model assumes that the equity markets efficiently price both the market consensus on the AI-outlook and the volatility surrounding it. The volatility channel is non-linear where it counts: the same shock produces larger spread reaction for names with currently lower volatility (e.g. Apple and Nvidia), and less spreads reaction for names already has high equity volatility (e.g. Oracle). Rates stress is less of a concern, as a 1pp move in rates only impacts distance-to-default by 0.07 standard deviations – a reasonable result while debt remains modest and off-balance sheet commitments lack rate sensitivity. A simultaneous stress scenario – equity down 10%, volatility up 10pp, plus debt up 50% (keeping the high issuance pace) – costs the panel two standard deviations of distance-to-default on average, which implies spread widening of around 20bps. Recognizing all the uncommenced leases pushed distance-to-default another 0.71 standard deviations shorter or around 10bps. Risk is firmly skewed to the downside in credit.

Debt and equity conditions are not two separate risks here – both enter distance-to-default, one raising the barrier and the other the cushion. Today they pull in opposite directions. Debt has grown hard on both sides of the disclosure line, yet distance-to-default has held for most of the big tech companies, because the equity leg moved the other way: market capitalizations are 40–70% above mid-2022 and volatility for all of them except Oracle sits at 24–39%, compared with the 41–57% of 2022. The equity market has also already taken sides: Apple and Nvidia re-rated through 2026, while Microsoft, Meta and Oracle de-rated, so the cushion is thinning precisely where the commitments are largest. For now spreads are held tight by the price-insensitive demand we described in July. But the offset is one-directional: commitments migrate to the financial statements on schedule regardless, so a turn in the volatility regime would raise the vulnerability and thin the cushion at the same time. In a market with less tailwind from demand that repricing tends to arrive quickly.

Table 3: Sensitivity and stress test – model-implied distance-to-default change by scenarios (based on Mid-2026, excluding off-Balance sheet debt)

Scenario	Average change (standard deviation)
Equity vol +1pp	-0.20
Equity vol -1pp	0.21
Equity price +10%	0.11
Equity price -10%	-0.13
Debt level +10%	-0.11
Debt level -10%	0.13
Risk-free +1pp	0.06
Risk-free -1pp	-0.06
Horizon T +1y	-0.42
Horizon T -1y	0.58
Stress scenario (price -10%, vol +10pp, debt +50%)	-2.00
Off-BS debt included	-0.71

Sources: Bloomberg, company filings, Allianz Research. Cells show the change in model-implied physical distance-to-default (standard deviations) from the mid-2026 baseline, one parameter at a time except in the combined-stress row.

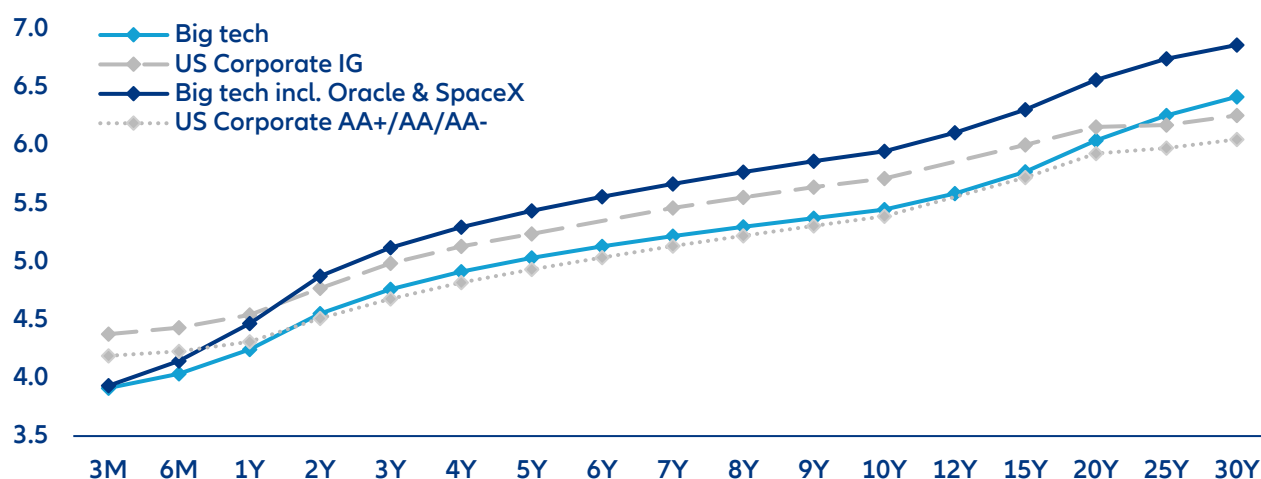
What this means for credit investors

First and foremost, based on today's balance sheets at the big-tech companies we examined, imminent default risk is remote for most of the panel. The finding is the dispersion and the direction. Their improvement in ranking compared with the broader market since 2022 came from the equity side – higher market capitalizations and lower volatility – and sits with the names that carry no commitment burden. Every name with large off-balance sheet liability deteriorates once that is counted. And the newly liquid CDS market has begun siding with the model approach, e.g. Meta CDS tends to reflect part of the off-balance sheet commitments. Default risk is still remote, but spread risk will increasingly surface.

Second, the metric to monitor is debt migration. The disclosed commitments commence and will move to financial statements between now and roughly 2031. The liability leg is contractual and arrives on schedule, while the equity leg reflects a market expectation that can reverse in a quarter, given uncertain monetization, or a technology jump to more efficient chips. The models say this will show up first as quality migration – e.g. one to two notches at Microsoft, Alphabet and Meta on full recognition, and if not matched by additional revenue – while spreads stay pinned until cushions are far thinner. The market pricing reflects the direction, but not yet the size: the 2026 widening covers part of the off-balance sheet liability, and the unpriced portion grows as new commitments are signed.

Third, after recognition of off-balance sheet liabilities, the main risk that remains is horizon and volatility, as Table 3 sizes it. A severe market repricing costs the panel two standard deviations of distance-to-default, or roughly 20bps of model spread. Recognizing the leases adds another 10bps on average. The two channels compound, because the commitments migrate on to the balance sheet on schedule, into whatever volatility regime prevails. The offset holding the rankings up is volatile: the equity leg can reverse in a quarter; the debt leg cannot. Horizon sensitivity concentrates the exposure in long-dated paper, consistent with the steeper curves on AI-linked issuance. In a repricing, the names with only a thin-cushion move by multiples, the rest by market beta. Oracle's CDS, already the sector's risk proxy, is where the move would be read first. Tech's curve steepness already prices the long horizon as the key risk, and that may extend to the front-end.

Figure 5: Big tech curves vs IG and AA benchmarks – higher premium at long-end



Sources: Bloomberg, Allianz Research

Tech has avoided the iceberg so far, but the navigation is getting trickier. The metric to watch is spread versus quality ranking within the corporate market, measured to include financial commitments and equity volatility. In recent years, the search for yield and tech's low initial weight in credit portfolios provided a tailwind that may turn into a headwind. With technology accounting for an ever-larger share of credit, integrating this into position sizing and maintaining the flexibility to adjust course if the (A)Iceberg appears will be key.

These assessments are, as always, subject to the disclaimer provided below.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

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Sector Atlas 2026:

The three-speed economy: How corporates navigate AI, geopolitics and fragmentation

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Executive summary



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- A challenging but still holding macroeconomic backdrop.** Global growth is expected to ease to +2.5% in 2026 before rebounding to +2.9% in 2027, propped up significantly by AI investment, which alone contributes roughly a third of US growth. Strip that pillar out and the divergence is stark: the US grows +2.1%, the Eurozone limps at +0.9% with almost no AI upside and China holds at +4.7% on exports and high-tech manufacturing despite weak domestic demand. A reloaded US trade war compounds the split even as Middle East de-escalation removes one acute risk while still leaving flare-up potential on the table. Across our sectors, this uneven backdrop is producing three distinct realities rather than one shared cycle.
- The semiconductor industry is riding the AI supercycle while other tech-related sectors face mixed outlooks.** Hyperscaler infrastructure spend is expected to reach USD725bn in 2026 (+80% y/y) and above USD1trn in 2027, propelling global chip sales toward USD1.5trn. As a result, semiconductor sales growth stands at about +90% thanks to the rise in memory price. The picture is mixed for other tech-related sectors: Software & IT services demand remains solid (+3.6%) but investor confidence has cracked, with leading names down 25-30% on GenAI-erosion fears. For computers and smartphones, AI is a cost story as component shortages are compressing margins while PC and smartphone volumes fall 11-14%, while telecom services benefit via data-center interconnect but are exposed to any pullback in AI capex, and to LEO-satellite competition eroding high-margin niches.
- Cyclical sectors are absorbing higher rates, geopolitical turmoil and a reloaded trade war.** A number of sectors are facing still-restrictive financing costs, tariffs and thin pricing power that forces companies to absorb rather than pass through input-cost inflation. The automotive sector has moved from unit volume to software and data monetization, with Chinese EV competition especially hurting Europe. Construction remains rate-sensitive, with residential and commercial demand still soft, offset mainly by data centers. Retail, textiles and household equipment are seeing consumers trading down, keeping headline sales resilient but eroding gross margins underneath. Transportation remains in flux, with shipping turbulence in the Middle East and high oil prices. Transport equipment and machinery & equipment are benefiting from defense spending but the recovery is gradual and vulnerable to a renewed growth or trade shock. In the chemicals sector, Europe's energy-cost disadvantage, sharpened by the Middle East conflict's exposure of import-dependent feedstock flows, is turning a cyclical downturn into a lasting divide between US/Middle East cost leaders and higher-cost European producers. Lastly, the metals sector is benefiting in segments close to copper, lithium and rare earths (with strong demand tied to electrification and AI infrastructure) while iron & steel remains challenged.

- **Traditional defensive sectors have different risk profiles in the current context.**

Pharma still benefits from strong patents and pricing power, while its US/Europe innovation base largely insulates it from the turbulence hitting everything else. In the energy sector, oil & gas players, especially US producers, are benefiting from higher oil prices but Middle Eastern, Asian and European players face a number of challenges, from sourcing feedstocks to reaching customers. Renewables might benefit in the current context of heightened focus on energy sovereignty and resilience.

- **Earnings remain resilient, but growth is increasingly concentrated in AI, defense and energy, while cyclical sectors continue to face significant headwinds.** In the US, revenue growth reached +15.2% y/y, its strongest pace since Q2 2022. Semiconductors were the standout, supported by the AI investment supercycle, while energy benefited from higher oil prices. Metals & mining also improved sharply, although gains largely reflected tariffs, stronger commodity prices and policy-driven margin expansion rather than broad-based demand. Chemicals showed signs of recovery, but underlying conditions remain fragile. Airlines were a clear weak spot as higher fuel costs drove a -44% y/y EPS decline. Europe delivered similarly strong headline revenue growth (+11.2%), but performance was concentrated in energy, defense and semiconductors. Automotive, transportation, discretionary retail and metals remained under pressure amid weak demand, high financing costs and global competition. Chemicals rebounded as geopolitical disruptions temporarily improved pricing power, though Europe's structural energy-cost disadvantage remains.

Country and Sector Matrix

		Automotive manufacturers	Automotive suppliers	Construction	Transportation	Chemicals	Pharmaceuticals	Agri-food	Textiles	Paper	Electronics	Metals	Retail	Machinery & Equipment	Transport equipment	Software & IT Services	Household equipment	Computers & Telecom	Energy
North America	US	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Canada	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Latin America	Brazil	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Mexico	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Argentina	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Colombia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Peru	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Ecuador	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Uruguay	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Paraguay	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Panama	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Chile	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Costa Rica	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Dominican Republic	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Guatemala	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Western Europe	France	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Germany	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Spain	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Italy	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	UK	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Netherlands	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Switzerland	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Sweden	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Norway	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Belgium	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Austria	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Denmark	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Greece	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Finland	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Portugal	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Ireland	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Luxembourg	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Turkey	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Poland	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Czech Republic	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Central & Eastern Europe	Romania	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Hungary	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Croatia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Bulgaria	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Slovenia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Lithuania	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Latvia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Estonia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Slovakia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Algeria	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Morocco	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Saudi Arabia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	UAE	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Africa & Middle East	Israel	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Qatar	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Kuwait	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Oman	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Bahrain	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	South Africa	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Japan	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Australia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	China	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	India	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Asia Pacific	Indonesia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Malaysia	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Singapore	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Philippines	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Korea (South)	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	New Zealand	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Thailand	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Taiwan	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Vietnam	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
	Hong Kong	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

● Low risk: sound fundamentals; very favorable or fairly good outlook.
 ● Medium risk: signs of weaknesses; possible slowdown.

● Sensitive risk: structural weaknesses; unfavorable or fairly bad outlook.
 ● High risk: imminent or recognised crisis.

Agrifood

Resilience tested by climate, trade fragmentation and input costs

Sector rating

M Sensitive risk for enterprises

Strengths & weaknesses



- Fundamental demand resilience: food is non-discretionary, and rising populations - particularly in emerging and middle-income economies; underpin durable long-run volume growth.
- Pricing power: despite elevated input costs in recent years, agrifood companies have largely maintained pricing power, keeping gross margins broadly stable and demonstrating the sector's ability to pass through cost pressures.
- Technology adoption is structurally improving yield efficiency, traceability and supply-chain resilience.



- Climate disruptions and extreme weather events are increasingly frequent and severe, causing yield volatility, commodity price spikes and infrastructure damage across major producing regions.
- Input cost pressures from fertilisers and energy to logistics and packaging remain structurally elevated, compressing margins for producers operating on thin spreads.
- Supply-chain vulnerability is high: the sector is simultaneously exposed to policy shocks (export curbs, trade barriers, subsidy changes), disease outbreaks and geopolitical disruptions to key commodity corridors.
- Labour shortages persist in key agricultural regions due to urban migration and ageing demographics, and stricter immigration policies in major economies risk intensifying near-term capacity constraints.

Sector overview

What to watch?

- Trade policy and food security: Export curbs, import tariffs and bilateral trade agreements are reshaping commodity flows globally. Governments are increasingly treating food supply as a strategic priority, introducing compliance costs and market distortions that are likely to persist.
- Climate and crop cycles: Cereal production has been high in 2025-2026, but climate-driven variability in key growing regions remains a major source of year-to-year revenue uncertainty.
- Labor and automation: Filling workforce gaps in farming and food processing through automation and robotics is a multi-year transition. Companies investing early in precision agriculture and harvest robotics will gain durable competitive advantages.
- Technology and biotech adoption: Advances in synthetic biology, AI-driven agronomy and alternative proteins are gradually reshaping what the sector produces and how, with adoption rates in emerging markets accelerating.

The global agrifood sector is one of the world's largest economic systems, employing approximately 1.2bn people and producing nearly 11bn tons of food annually. The market encompasses upstream agriculture (cereals, oilseeds, livestock, aquaculture), food and beverage processing and a rapidly expanding agritech layer spanning digital farming tools, biotech inputs and smart logistics. Asia-Pacific accounts for the largest share of production, while demand growth is increasingly driven by expanding middle classes in South and Southeast Asia, Africa and Latin America.

Global gross output for upstream agriculture (crops, livestock, fisheries) is estimated at approximately USD7.6trn in 2025, rising to around USD8.2trn in 2026 – a gain of roughly 8%, with a 2026-2028 CAGR of approximately +3.6%. Food, beverage and tobacco processing is a separate, larger stage of the value chain, at approximately USD9.3trn in 2025 rising to around USD10trn in 2026 (+8.6%), with a 2026-2028 CAGR of approximately +5%. However these two figures should not be summed, since processing output already embeds upstream agricultural output as an intermediate input. This growth reflects a combination of stable to firming commodity prices in most categories and continued volume growth in beverage sub-sectors. Food price indices in 2025 were higher than in 2024 but remain approximately 21% below the March 2022 peak. Cereal markets are well-supplied (2025 production reached a record) keeping staple prices broadly in check, while meat and dairy prices remain elevated due to persistent supply-side constraints.

Profitability varies significantly by value-chain position. Farming/Agriculture has an EBITDA margin of approximately 8.0% and a net margin of approximately 4% - thin margins typical of a commoditized, price-taking segment with real but limited pricing power. Food Processing shows a similar EBITDA margin (approximately 15.3%) but a lower net margin (approximately 3%), reflecting high input-cost pass-through. Food wholesalers/distribution is the thinnest of all (EBITDA margin approximately 3.7%, net margin approximately 1%), consistent with a low-margin, high-volume logistics function. Beverages show the strongest and most differentiated profitability: Beverages (alcoholic) carry a EBITDA margin of approximately 29.5% and beverages (soft) approximately 22.8%, both well above the food-processing average, which

supports the observation that beverage sub-sectors are outperforming on both growth and margin. As a general pattern, margins expand moving downstream from farming toward branded, processed and beverage products, while distribution and wholesaling remain structurally low-margin regardless of the underlying commodity cycle.

The key risk factors for the sector over the medium term are structural rather than cyclical. Climate change is increasing the frequency of extreme weather events that disrupt yields and logistics in ways that are difficult to plan around. Trade-policy fragmentation – including agricultural tariffs, export bans and shifting bilateral agreements – is adding cost and complexity to global supply chains. Labor availability in key producing regions remains a constraint, and the structural solution (automation) is a multi-year investment cycle. Companies that invest in supply-chain diversification, precision agriculture and technology adoption will be better positioned to sustain margins and manage volatility than those relying on traditional sourcing models.

Subsectors

Upstream agriculture: Growers and producers of foundational agricultural goods including cereals, oilseeds, fresh fruits, vegetables, livestock and aquaculture. Performance is closely tied to weather patterns, commodity price cycles and input cost inflation.

Food and beverage processing: Firms that convert raw agricultural commodities into packaged, shelf-ready food and drink products. Encompasses processing, packaging and logistics, with beverage sub-segments demonstrating stronger recent revenue growth and pricing resilience.

Agri-tech and biotech: A cross-cutting segment leveraging digital innovation (AI, IoT, drones, robotics), synthetic biology and precision farming tools to improve yield, traceability and sustainability across the upstream and downstream supply chain. The fastest-growing part of the broader agrifood ecosystem.



Automotive

Battling a new business environment and technology shift

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Stronger demand for EVs in Europe aligns with tightening framework, pushing for a further electrification of new supply
- Strategic sector benefiting from regular state support / protection
- Well-developed and competitive ecosystem favoring innovation and cost control
- New and lucrative service-derived revenue channel under rising collection of data
- Older fleet in Europe (high potential for renewal)



- Strong competition from Chinese OEMs orienting retail car price to the downside
- Reliance on China for software-defined tools, sensors and batteries.
- Increasing tariff and non-tariff barriers altering trade exchange volume and profitability
- New minimum domestic sourcing criteria in
- Low ROI for autonomous mobility R&D (weak demand for that technology)

Sector overview

What to watch?

- Tariff and non-tariff barriers and consequences on the supply chain
- Adoption pace of EVs and change over national demand-supportive policy
- Cost-cutting programs from large western OEMs and impacts on local/regional ecosystems
- Evolution of the China's market share abroad (Europe & Asia)
- Impacts of new regulatory framework on part sourcing on European/US automotive ecosystems
- New range of services derived from a broader adoption of software-defined technology inside vehicles
- -Development of robotaxi services across the US and Europe and evolution of demand for autonomous vehicles

Demand is not the sector's problem, the mix is. EU registrations rose +6% YTD in H1 2026, with Battery Electric Vehicles (BEV) at 21% versus roughly 16% a year earlier, and now neck-and-neck with Internal Combustion Engines (ICE), whose market share sharply decline in one year from 28% to 22%. Hybrids remain the dominant segment, with a 37% market share, slightly up by 2pps comparatively to the same period last year. New geopolitical headwinds in the Middle East and increasing energy volatility have not altered demand despite energy volatility. In fact, they seem to have rekindled new interest for new and non-fossil engine models as operating and maintenance costs have sharply increased since 2022, especially on the back of an ageing European fleet. The US is moving the other way: The EV share fell from 10.5% to 5.9% after the credit lapsed, with 2026 sales set to decline 3% to 15.8mn units. Directionally, every unit of growth is dilutive – volume shifts toward lower-ASP mass-market EVs and incentive-dependent segments, so revenue holds while contribution margin erodes. The commercial opportunity sits where volume and margin still coincide: corporate fleets, around 60% of European new car sales and the next policy lever, plus aftermarket and service revenue on an older fleet – the one reliably high-margin annuity in the model.

Low pricing power amid fierce competition from new entrants. Pricing power has passed to the low-cost entrant. Chinese brands reached a record 10.9% European share in June and are gaining share over historic foreign competitors. The financial consequence is direct: with the price floor exogenous, European groups cannot recover cost inflation through pricing, which is why margin guidance has compressed to a low single-digit rate (1-5%) and is today oscillating near the low seen during the crisis period. In China, EVs are ~55% of sales and 70% of BEVs already undercut the average combustion car – the offshore profit pool is structurally impaired, not cyclically weak. In this new environment, assuming permanent price compression in mass-market segments, carmakers have no choice but to defend margins through mix, cost and services rather than volume recapture. This is one key challenge European carmakers imperatively need to address this year.

Trade policy and sourcing rules: footprint, not compliance, is the adjustment. Tariffs are now priced as permanent, and that assumption is what forces irreversible capital decisions – the remaining duty architecture survived the Supreme Court striking down the IEEPA tariffs. The result is industrial migration in both directions: a dozen US EV models were pulled because imports could not clear 25% duties and 100% levies on Chinese-built cars profitably, forcing domestic assembly or exit. Meanwhile Chinese groups build European plants that bypass EU duties entirely – protection relocates the competitor inside the wall rather than keeping it out, and defense shifts from trade policy to domestic cost. Sourcing rules are the harder lever: The EU is trading compliance flexibility for “made in the EU” conditionality, a Battery Booster Strategy and fleet greening, with residual 2035 emissions offset through EU-made low-carbon steel and e-fuels – an explicit attempt to unwind dependence on Chinese cells, sensors and SDV tooling, backed by EUR108bn committed to European gigafactories targeting ~1.4 TWh by 2030, though LFP cost leadership stays Chinese.

A whole ecosystem fragilized by industrial revamping and liquidity constraints. Restructuring is the dominant use of cash and consumes before it releases – a multi-billion write-down on overhauling cost-cutting programs and downsizing line-up. OEM liquidity remains adequate – net cash positions and captive finance arms – and investment-grade solvency is not in question at the assembler level. The fragility is one tier down. 76% of European suppliers expect margins below 5% in 2026, beneath the threshold needed to fund innovation,

after the supply chain shed roughly 90,000 jobs (net) in two years. Weak margins combined with leverage and mandatory electrification capex is an insolvency profile, not a cyclical squeeze. The main risks for the industry are rather in supplier distress ahead of OEM credit metrics. Indeed, a disorderly failure in tier-2/tier-3 transmits into production stoppage and a working-capital shock far faster than any demand shortfall, and it is the most plausible route to a 2027 downgrade cycle.

Software and autonomous mobility: the model shifts from unit sale to lifecycle annuity. The strategic prize is no longer the vehicle margin but the recurring stream behind it – paid-in functions activated post-delivery, data monetization and eventually autonomous mobility services – turning a one-off transaction into cash generation across the ownership cycle and structurally lifting the revenue quality of a business currently valued on volume. That transition is capital-hungry before it pays and we see no meaningful P&L contribution inside the 12-24M forecast window. The deeper constraint is ecosystem: few European or US players sit at the leading edge of IoT, embedded software and high-end sensing, where Asian suppliers dominate, so the pivot to services runs on imported critical content. The new “made in the EU” sourcing conditionality is designed precisely to seed a domestic ecosystem around this technology shift, but the interim is disruptive on both sides of the value chain: OEMs absorb new cost lines and heavy organizational change – moving from mechanical engineering to software release cycles – while suppliers face the harder adjustment, competing against entrenched foreign incumbents while re-tooling for a demand profile they were not built to serve.

Subsectors

Manufacturers: This includes firms that assemble all the parts to produce a vehicle and oversee second-hand distribution and marketing through their own channels or via an external network of suppliers.

Suppliers: This includes all firms that collaborate with manufacturers by supplying them raw materials, parts, tyres, batteries, software...etc that will be incorporated in the final composition of the vehicle.

Retailers: This includes firms that buy cars from OEMs before redistributing to the retail market.



Chemicals

An uneven recovery

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Diversified end-markets and broad product portfolios provide some resilience
- Strong competitive positioning in higher-value specialty segments.
- Structural growth opportunities linked to sustainability and the energy transition.
- Strong innovation capabilities and technological expertise.
- Global scale, integrated operations and high barriers to entry.



- High capital intensity can weigh on returns during periods of weak demand.
- High exposure to energy and raw material price volatility.
- Cyclical demand, overcapacity and margin pressure as many chemical products are closely tied to global economic activity and industrial cycles.
- Significant ESG, decarbonization and regulatory transition challenges: Meeting increasingly stringent climate, safety, and chemical regulations requires major investment and operational transformation.
- Long-term disruption risk from changing materials demand and the energy transition

Sector overview

What to watch?

- Geopolitical fragmentation and supply-chain realignment: Persistent geopolitical tensions, trade restrictions and uncertainty and regionalization trends are reshaping chemical supply chains, accelerating the shift toward local production, strategic sourcing and supply diversification.
- China's evolution from growth engine to competitive disruptor: China remains the world's largest chemicals market and producer, but slower domestic growth (GDP: +4.7% in 2026 and +4.4% in 2027) and continued capacity expansion are increasing global competition, putting pressure on pricing and reshaping trade flows.
- Trade policy and industrial protectionism: Changing tariffs, export controls/bans and industrial policies in major economies could increase input costs, disrupt global trade patterns and accelerate the localization of chemical production.
- Energy competitiveness and regional cost dynamics: Differences in energy prices and feedstock availability will remain a key determinant of competitiveness, particularly as

European producers continue facing higher structural costs compared with regions benefiting from cheaper energy and raw materials.

- Capital allocation toward innovation and efficiency: Leading chemical companies are expected to prioritize investments in advanced materials, automation, artificial intelligence and process optimization to improve productivity, develop differentiated products and strengthen competitive positioning.
- Decarbonization as both a challenge and growth driver: Regulatory pressure and customer demand are accelerating investment in low-carbon technologies, circular solutions, bio-based chemicals, sustainable packaging and materials supporting electrification and the energy transition.
- Industrial policy and the reshoring of strategic industries: Government incentives supporting domestic manufacturing, supply-chain resilience and strategic sectors will influence future chemical capacity investments, particularly in regions seeking to reduce dependence on foreign suppliers.

The global chemicals industry remains one of the most strategically important industrial sectors, underpinning virtually every major value chain from construction and automotive to electronics, agriculture, healthcare and consumer products. The sector encompasses a broad range of segments, including basic chemicals, petrochemicals, polymers (plastics), packaging, specialty chemicals, agrochemicals (fertilizers and pesticides), coatings and pharmaceutical intermediates. While long-term growth fundamentals remain intact due to rising material demand in emerging economies, technological innovation and the increasing use of advanced materials, in 2026 the industry is navigating a widening structural divide between globally competitive producers and higher-cost regions.

Asia-Pacific, led by China, continues to dominate global chemical production and consumption, accounting for approximately half of global output and demand. China has expanded capacity aggressively over the past decade, particularly in petrochemicals, polymers, and basic chemicals, supported by large-scale industrial policy, integrated refining and chemical complexes and strong domestic demand. However, the rapid build-out of Chinese capacity has contributed to global oversupply in several commodity segments, placing pressure on margins worldwide and accelerating rationalization among higher-cost producers. At the same time, India continues to show strong growth, supported by domestic demand and specialty chemicals. Across APAC, companies are increasingly prioritizing higher-value specialty chemicals, advanced materials and sustainability initiatives to offset pressures in traditional commodity markets.

The US remains one of the strongest global competitors in 2026, benefiting from abundant shale gas resources, lower-cost ethane feedstocks and highly integrated petrochemical infrastructure along the Gulf Coast. These structural advantages have strengthened the competitiveness of US producers, particularly in ethylene, polyethylene and other olefin-based products.

Europe, by contrast, continues to face a challenging industrial outlook, with its chemical sector undergoing a significant structural adjustment as high energy costs, carbon pricing, regulatory complexity and weak demand undermine the economics of energy-intensive production. Despite some improvement in business confidence, the recovery remains fragile amid persistent geopolitical and

trade uncertainties, while the industry's performance remains well below pre-crisis levels. Europe's structural energy cost disadvantage – particularly natural gas prices that remain significantly above those in the US – continues to erode competitiveness, discourage investment and accelerate the relocation or closure of production capacity. These challenges are especially evident in olefins, the essential building blocks for plastics and many industrial materials. Europe has shifted from being broadly self-sufficient in polyolefins to becoming increasingly dependent on imports, a trend expected to continue over the coming decade as domestic olefin capacity declines. This erosion of production capacity extends beyond the chemical sector, posing broader risks to industrial competitiveness and supply-chain resilience, given the strategic importance of olefins and their derivatives across manufacturing value chains. On top of this, the European chemical sector faces a strategic dilemma: balancing decarbonization objectives with the need to preserve industrial capacity. The region's transition toward lower-carbon production is necessary, but the current cost environment creates significant challenges for energy-intensive industries. Electrification of chemical processes, particularly steam crackers, represents one potential pathway. Pilot projects demonstrate that electric crackers could significantly reduce emissions, potentially cutting carbon intensity by up to 90% in certain applications. However, commercial deployment requires substantial investment, competitive electricity prices, reliable power supply, supportive policy frameworks and regulatory certainty.

Geopolitical risks and supply-chain security issues have intensified in 2026. Geopolitical developments are becoming an increasingly important factor for chemical markets. The Middle East conflict has highlighted the vulnerability of global material supply chains and reinforced the strategic importance of domestic chemical capacity. Disruptions to energy flows can quickly affect petrochemical margins, benefiting low-cost producers while increasing pressure on regions dependent on imported energy and feedstocks. In 2026 and 2027, chemical companies are likely to continue reassessing supply chains, prioritizing resilience alongside cost efficiency. This could accelerate regionalization trends, encourage investment in integrated production hubs and increase strategic importance of countries with secure energy and feedstock access.

Entering 2027, the sector recovery should be highly uneven across regions and segments. With global GDP anticipated to improve in 2027 (+2.9% from +2.5% in 2026), industrial activity should stabilize next year, with chemicals demand growth expected to improve gradually. However, the industry will increasingly be defined by a widening gap between producers with structural cost advantages (US and Middle Eastern companies) and those burdened by higher energy and feedstock costs, with European players expected to remain at a competitive disadvantage. The survival of Europe's chemical industry will depend on its ability to preserve profitable capacity, accelerate in low-carbon technologies, improve its industrial infrastructure and create a more predictable investment environment. Commodity chemical markets are likely to remain primarily driven by cyclical pricing dynamics, limiting the scope for structural differentiation, while specialty chemicals and advanced materials are expected to benefit from stronger pricing power and long-term demand trends.

Subsectors

Basic or commodity chemicals: Chemicals substances used as a starting material for the production of a wide variety of other chemicals. Examples include chlorine (used as disinfectant and for water treatment), sulfuric acid (used in metallurgy, for refining petroleum products or production of explosive materials), vinyl chloride (used to produce PVC, for wall coverings, housewares and automotive parts), aluminum sulfate (used for water treatment, in agriculture and in paper production), sodium carbonate (used in the manufacture of detergents, soaps and paper), acetone (commonly used in pharmaceuticals), titanium dioxide (used in the cosmetic and food industries), among others.

Under this sub-sector, we can find another distinction – Petrochemicals - when looking to the origin of the raw material. Petrochemicals are also organic chemicals, but they are made from crude oil and natural gas. Examples include olefins, methanol, butadiene, benzene, ethylene glycol, polyethylene, etc. Used as raw materials in the manufacture of polymer products like plastic, detergent, adhesive, rubber, tires, food packaging, and elastic bands.

Specialty chemicals: Specialty chemicals are a range of compounds that are produced in smaller quantities, when comparing to basic chemicals, as they are high-value products that are sold based on functionality and following certain formulations that make them “special”. Under this sub-sector we can find a variety of classification according to the end-market, such as antibiotics, adhesives, pesticides, fertilizers, cleaners, inks, paints and coatings, fragrances, chemicals for the food & beverage industry (food additives and flavors), etc. In terms of commercial applications, producers within the specialty chemicals sub-sector cater to the needs of their customers on an individual level (production is tailormade under certain specifications), and because of this these chemicals can be sold at very high prices.



Construction

Execution and financing challenges persist

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Robust infrastructure investment pipelines and strong project backlogs provide multi-year revenue visibility.
- Strong demand from high-growth segments, including data centers, logistics, semiconductors and energy infrastructure.
- Structural demand driven by aging infrastructure, urbanization and population growth.
- Energy transition accelerating investment in renewable energy, transmission grids, storage and related infrastructure.
- Stabilizing construction material and input costs supporting margin recovery and project profitability.
- Increasing adoption of digital construction technologies and industrialized building methods improving productivity and execution.



- High interest rates and tighter financing conditions constraining activity in interest-sensitive segments, particularly residential and commercial construction.
- Persistent labor shortages and skills gaps continue hampering construction globally, while tighter immigration policies in many economies are also weighing on the sector.
- Low productivity and slow adoption of innovation limiting efficiency gains and margin expansion.
- Complex regulatory frameworks, permitting delays and compliance requirements increasing project execution risk.
- High project execution risk due to cost overruns, schedule delays and fixed-price contract exposure.
- Earnings volatility driven by cyclical demand and exposure to macroeconomic conditions.

Sector overview

What to watch?

- **Financing costs and interest rates:** The pace of monetary policy easing will be a key determinant of construction activity. Lower interest rates could support a recovery in residential and commercial construction, while prolonged restrictive financing conditions would continue to weigh on private investment.
- **Public policy and infrastructure spending:** The pace of implementation of major public investment programs – including the US Infrastructure Investment and Jobs Act (IIJA), Europe's NextGenerationEU funds and infrastructure initiatives across Asia and the Middle East – will shape sector growth and order pipelines.
- **Labor availability:** Persistent labor shortages, aging workforces and tighter immigration policies in several economies will remain key constraints on project execution, costs and delivery timelines.
- **Sustainability and decarbonization:** Tightening environmental regulations, stricter building standards and increasing demand for low-carbon materials and energy-efficient buildings will continue to reshape investment priorities and competitive positioning.
- **Input costs and supply chains:** Although material cost inflation has moderated, supply-chain disruptions, energy prices, tariffs and geopolitical tensions remain key risks for project costs and scheduling.
- **Data center and energy infrastructure investment:** Continued investment in AI-driven data centers, electricity grids, renewable energy and transmission infrastructure is expected to remain one of the sector's strongest growth drivers, partially offsetting weakness in more cyclical end markets.

The construction sector is a highly cyclical industry with significant exposure to macroeconomic conditions, government spending, financing costs and commodity prices. However, its end-market diversification – including infrastructure, residential, commercial, industrial and energy projects – provides varying degrees of resilience across market cycles. As it is highly competitive, the sector is characterized by low operating margins, execution-intensive projects and elevated working capital requirements, making project selection, cost control and cash conversion critical differentiators. While public infrastructure spending remains the main driver of long-term growth in developed markets, private construction activity is more sensitive to interest rates and business confidence. Geographic dynamics also differ markedly, with North America benefiting from large-scale infrastructure and AI-related investment (data centers); Europe supported by public infrastructure, renovation and energy-transition programs; emerging markets driven by urbanization and infrastructure deficits and China continuing to face structural weakness in real estate and construction activity.

The global construction sector has remained resilient throughout 2026, supported by infrastructure spending and sustained investment in strategic sectors, despite weaker private construction activity. Public infrastructure, energy-transition projects and digital infrastructure continue to underpin growth, while residential and commercial construction remain constrained by still elevated financing costs and cautious private investment. Infrastructure has consolidated its position as the sector's strongest segment, driven by government investment in transportation, electricity grids, water systems, renewable energy and defense. At the same time, AI-driven demand for data centers has emerged as one of the fastest-growing construction markets globally, creating significant opportunities for engineering and specialist contractors. By contrast, residential activity is recovering only gradually from a low base, while commercial real estate continues to be affected by higher financing costs and structural changes in office demand.

Regional dynamics remain uneven. North America continues to outperform, supported by large-scale public infrastructure programs and stronger private investment in energy and data centers. Europe is experiencing a gradual recovery as infrastructure spending, energy-transition investment and renovation activity somewhat offset subdued private building construction, although growth remains stronger in Eastern Europe than in Western markets. Emerging economies – particularly India and the Middle East – continue to benefit from rapid urbanization, infrastructure investment and favorable demographic trends. Conversely, China remains the weakest major market as the prolonged downturn in the property sector continues to weigh on construction activity despite targeted government support for infrastructure.

The sector's operating environment has improved compared with the inflationary pressures experienced in recent years. Construction material inflation has largely stabilized, easing pressure on project margins and improving bidding conditions. Nevertheless, labor shortages, skilled workforce constraints and permitting delays remain persistent challenges across many markets, limiting execution capacity and increasing project risk. Given the sector's traditionally low margins and project-based business model, disciplined contract selection, cost control and efficient project execution remain essential to sustaining profitability and cash generation.

Looking ahead to 2027, the global outlook remains constructive, although growth is expected to continue varying across end markets. Civil engineering is likely to remain the strongest-performing segment, supported by continued investment in energy infrastructure, electricity transmission, transportation, water infrastructure and defense projects. Data-center construction is expected to remain a major structural growth driver in some countries as hyperscale cloud providers and AI-related investment continue to expand globally. Residential construction should take more time to recover as financing conditions ease, although activity is expected to remain below previous cycle peaks in several markets, while commercial construction is likely to experience a more modest recovery.

Subsectors

Residential construction: This segment includes the construction of new homes and apartments, as well as the renovation and repair of existing ones.

Non-residential construction: The construction of commercial buildings (offices, retail), industrial buildings (factories, warehouses, data centers) and public buildings (schools, hospitals etc.).

Infrastructure construction: Construction of large-scale public projects such as highways, bridges, railways, airports and utilities.

Construction services: Services related to construction, such as architectural and engineering services, project management etc.



Electronics

The heavy price of AI

Sector rating

M Medium risk for enterprises

Strengths & weaknesses



- Moderate pricing power for premium brands (pass-through abilities when costs rise)
- High fleet renewal potential under broader AI adoption and usage from B2C/B2B
- Strong ecosystem of suppliers (rapid & qualitative execution, flexibility)
- Slow to mid capex intensity (depending on automation level into production line)
- Easily automated and transportable activities if market conditions change



- High exposure to chip shortage (higher costs = margin squeeze for medium to low-end products)
- Mild consumer demand amid slow technology upgrade cycle on telco hardware products
- Highly concentrated supply chain in Asia (exposure to export controls & targeted tariffs)
- Consumer confidence dependent (higher interest rates to sap demand for non-necessary durables)
- Labor-intensive business easily replicable (fragmented competition = thin margin for assemblers)

Sector overview

What to watch?

- Evolution of production cost line through memory chip shortages and potential impacts of broad-based margin squeeze on market concentration structure
- B2C demand for new AI-powered hardware in an inflationary environment
- B2B AI adoption and interest rate level impacts on IT hardware budget
- New growth channel explored by OEMs as top-line booster (leasing, paid-in services...)
- Offshoring and/or nearshoring investments amid stronger pressure to reduce China's footprint on the supply chain and disruption risks for global output.
- Integration of AI technology into manufacturing and forecasting processes to cut costs.



A large costs wave to tame. Underlying demand is saturated and now contracting under price pressure. Both core categories are set for their steepest drop in over a decade in 2026 – PC shipments down roughly 11% and smartphones down 14% – as rising prices push buyers to defer, lengthening PC lifetimes by 15–20%. The contraction is highly uneven: the entry level is collapsing (sub-USD500 PCs and sub-USD200 handsets losing ~20% of volume), while premium demand above USD800 is largely insulated. The mix is therefore shifting hard toward premium, where pricing power and ecosystem lock-in let vendors pass costs through, and toward the one growth pocket – foldables, up ~20% and the only segment where units and premium pricing rise together. The swing factor is B2C and B2B AI-hardware demand – an AI-PC and AI-smartphone refresh could reset upgrade cycles, but in an inflationary, higher-rate environment that catalyst is real yet delayed, its timing the single biggest variable for the topline.

The “RAMageddon” is threatening low-end hardware models as profitability erodes dramatically. The profitability outlook in 2026/2027 is dominated by one input: the heavy price of memory semiconductors. The AI data-center boom is siphoning DRAM, NAND and mobile memory capacity out of consumer devices, and the cost transfer is severe: memory is set to peak near 23% of PC bill-of-materials, up from 16% in 2025, with one major maker citing memory at ~35% of laptop material cost, up from 15–18% a quarter earlier, and notebook build costs potentially rising ~40%. This removes vendors’ ability to absorb costs, forcing device prices up ~17% for PCs and ~13% for smartphones.

The margin consequence bifurcates the field: Premium and scale players with pass-through pricing and supply leverage protect margins and gain share, while budget-focused assemblers face outright margin destruction. Contract manufacturers – already operating at ~2–4% margins – are most exposed, since higher component prices inflate revenue without lifting assembly fees. The structural danger is that a permanent step-up in the cost base (memory plus tariffs) converts a cyclical squeeze into a solvency and liquidity problem for weaker, thinly-capitalized players, accelerating consolidation toward those who can fund working capital through the shock.

Ecosystem strength challenged by changing market conditions. The sector’s persistent structural weakness is where it sits in the value chain: it captures little of the value it assembles. Profit is hyper-concentrated in a single premium ecosystem player, while component suppliers take the rents and the rest of the field fights over commoditized volume. This is a labor-intensive, easily-replicated activity with fragmented competition and thin margins – precisely the conditions under

which a cost shock triggers shake-out rather than shared pain. Expect intensifying market concentration as margin and liquidity stress force weaker brands and subscale assemblers out, and as scale, procurement leverage and pricing power become decisive. The competitive advantage accrues to those who own an ecosystem (recurring services, accessories, software) or command the manufacturing scale to negotiate component allocation – everyone else is a price-taker on both sides. One structural response worth watching: OEMs exploring new topline channels beyond the box – device leasing, device-as-a-service and paid-in software/services – to add recurring revenue and defend margins as unit economics deteriorate.

A supply-chain transformation to monitor: de-risking China strategy to gain traction. The physical supply base is both a strength and the sector’s sharpest structural risk. Its strength is a dense, flexible Asian supplier ecosystem capable of rapid, high-quality, high-volume execution, and activities that are relatively easy to automate and relocate when conditions change. The weakness is concentration: assembly was historically ~90% China-based, and finished devices are physically traded and directly tariff-exposed – a step-change

Subsectors

Electronic components can be loosely divided into **semiconductor** designers, **manufacturers** and companies specialized in assembly and **testing services**.

Semiconductors serve different purposes (computing, storage, power management, communication etc.) and are nowadays found in a large range of products used on a daily basis, such as cars, electronic devices, electric home appliances and video displays. More recently, the component has also been integrated into other “non-conventional” industries like agriculture, health care, transportation or utilities (grid network, waste management), creating a whole ecosystem based on data and connectivity features, known as the “internet of things” (IoT). Semiconductors are also a key component of artificial intelligence technology, which needs powerful calculation capacities for the training and inference of large language models (LLMs). Beyond chips, the electronic industry also includes manufacturers that produce everything from printed circuit boards to electronic displays through to resistors, capacitors and switches.

Energy

Fossil fuels resilient while renewables face a profitability reckoning

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Traditional oil and gas companies continue to generate robust free cash flows at current price levels, supported by production discipline and entrenched market positions. Strong balance sheets provide resilience against price volatility.
- Natural gas is increasingly positioned as a duration transition fuel, with major producers (particularly in the US, Qatar and Australia) expanding LNG export infrastructure to serve growing demand in Asia and Europe.
- Solar and wind remain on long-run cost-reduction trajectories through scale and technology innovation, and represent the cheapest source of new electricity generation in most markets.
- AI and data-center expansion is creating structural new electricity demand that reinforces the long-term investment case for all forms of power generation, including both renewables and gas.



- Renewables profitability remains structurally challenged: rising capital costs, grid interconnection bottlenecks and merchant power price risk are compressing returns on new renewable projects in many markets.
- Intermittency combined with inadequate storage capacity and grid infrastructure is limiting the effective contribution of wind and solar to system reliability, requiring costly backup capacity.
- Oil and gas face long-term demand uncertainty as the energy transition gradually erodes fossil fuel consumption in transport and heating, creating strategic planning challenges for capital-intensive upstream investments with 20-30-year time horizons.
- High financing requirements for both segments (capital-intensive renewables projects and new LNG and oil infrastructure) create sensitivity to interest rate levels and investor risk appetite.

Sector overview

What to watch?

- **Renewable deployment rates:** The pace at which wind and solar capacity is added will determine how quickly the energy mix shifts. China is leading installations while other major economies are scaling up but at slower rates constrained by grid and permitting bottlenecks.
- **Grid infrastructure investment:** In most major markets, grid expansion and modernization is lagging renewable capacity additions, creating a binding constraint on the clean energy transition and sustaining demand for grid infrastructure companies.
- **Peak fossil fuel demand:** Oil demand in transport is expected to peak within this decade in most scenarios. The timing and pace of this transition will determine how aggressively oil and gas companies need to diversify or divest.
- **AI and data-center power demand:** The energy requirements of AI training and inference represent a new and rapidly growing source of electricity demand that was not anticipated in most energy forecasts even a few years ago.
- **Geopolitical risks:** Supply routes for LNG, oil and critical minerals remain exposed to geopolitical disruption. Energy security concerns are driving both renewable diversification and domestic fossil-fuel expansion in different markets, often simultaneously.

The global energy sector encompasses oil and gas extraction and processing, power generation (fossil, nuclear and renewable), electricity transmission and distribution and the growing segment of grid-scale storage and smart infrastructure. Global gross output for utilities is estimated at approximately USD6.5trn in 2025, rising to around USD7.2trn in 2026, with a 2026-2028 CAGR of approximately +3%. Extraction (oil, gas, coal and other mining) is estimated at approximately USD4.8trn in 2025, rising to around USD5.6trn in 2026, though this segment is more cyclical and price-driven, with output projected to ease back toward USD5.5trn by 2028 as commodity prices normalize. These two figures span different value-chain stages (extraction feeds into power generation as an input).

The sector is also characterized by a pronounced structural duality. Oil and gas companies have enjoyed healthy cash flows in recent years, with Brent crude oscillating in the USD70–90 per barrel range – elevated enough to sustain profitability while not triggering the political pressure that accompanied the 2022 price spike. These companies have maintained capital discipline, returning cash to shareholders rather than pouring it into new upstream capacity, a stance that could constrain supply in the medium term. By contrast, the renewables segment has faced tighter margins and investor scrutiny as capital costs rose with interest rates and power-purchase-agreement prices proved insufficient to cover financing costs on new projects. The result is a slower-than-anticipated pace of renewable deployment in some markets, even as China continues to install capacity at a pace that exceeds the rest of the world combined.

As a result, profitability metrics differ substantially across energy subsectors. Oil & Gas Production and Exploration shows the strongest margins in the traditional fossil-fuel space, with an EBITDA margin of approximately 43% and a net margin of approximately 15%, benefiting from operating leverage at current price levels. Integrate oil & gas companies run thinner margins (EBITDA margin approximately 22%, net margin approximately 8%), reflecting the lower-margin downstream refining and distribution activities blended into their results. Utility companies show high and stable

margins (EBITDA margin approximately 35%, net margin approximately 14%), consistent with regulated, capital-intensive businesses with predictable cash flows. Green & renewable energy stands out for a large gap between operating and net profitability: its EBITDA margin is the highest in the group at approximately 58%, but the net margin is negative (approximately -11%), reflecting heavy depreciation and interest expense on project debt and impairments – direct evidence of the financing-cost and capital-intensity pressure.

The long-term direction of the sector is not in question: electrification of transport, heating and industry will drive substantial growth in electricity demand, and renewables will capture a growing share of that demand as costs continue to fall. The question is the pace of transition and the role that natural gas plays as a bridge fuel. The US has emerged as the world's leading LNG exporter, providing European and Asian buyers with an alternative to Russian pipeline gas and creating a new global gas trade architecture. AI and data-center energy demand represents an incremental demand layer that was not anticipated in earlier energy transition models and is reinforcing investment in all forms of reliable power generation – gas, nuclear and batteries – alongside renewables.

Subsectors

Fossil fuels: This includes oil, gas, and coal operations. Firms can be involved in upstream or downstream processes.

Renewables: Firms in the production/installation of renewable energy/projects.

Power: Firms in electricity generation, whether through fossil fuels, nuclear, solar, wind, hydro or other forms of renewable energy.

Grids and storage: This includes firms managing and building infrastructure needed to distribute and store energy.



Household Equipment

Repositioning toward services amid a stalled housing cycle and tariff pressure

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- AI development to create new demand for smart IoT products with new applications
- High entry barrier as business is capital intensive and certification-heavy
- Sustainability rules create new regular revenue stream via aftermarket services pool
- Steady margin rate for high/mid-premium segment as quality still remains a strong purchasing argument
- Diversified geographic revenue helps mitigate demand shocks as cycles diverge



- Sluggish recovery of housing market in western economies
- Supply-chain vulnerability (highly concentrated in Asia) favour new entrants amid manufacturers
- Longer product shelf life under new sustainability regulation
- Limited pricing power as demand is very elastic and customers keep price-seeking attitude
- Escalation of input costs due to tariffs and shortage of semiconductors

Sector overview

What to watch?

- Consumer spending momentum amid new inflationary pressures and higher rates
- Impacts on sales of pass-through policy applied to offset higher energy/software prices
- Consumer interest in new AI-embedded home equipment as potential booster of renewal cycle
- Any recovery signal on residential housing market, especially in China where demand remains deeply deteriorated
- Impacts of new tariff announcements on production costs and sourcing policy
- Development of aftermarket service activity as additional source of revenue
- Widening margin gap within the industry based on customer premium perception

A broken demand trigger as replacement cycle is lengthening.

The sector's difficulty is not weak appetite but a disabled purchase mechanism. Big-ticket appliance demand follows housing turnover and US existing-home sales remained sluggish through spring 2026, with the anticipated rate-driven rebound failing to arrive; Western Europe is flatter still, compounding at barely over 2% annually, with replacement cycles expected to engage during 2026 from a low base. Demand is also unusually elastic, with price driving consumer choice in most markets while quality arguments hold only in specific premium segments. China is the swing variable and currently a negative one – a subsidy program that pulled 2025–2026 demand forward into 2024 has created overhang, with domestic sales at risk of a significant drop absent extended trade-in support. As a matter of fact, we see no volume recovery in 2026/2027 without a housing turn.

Tariffs are becoming a working-capital problem. This is the acute pressure of the period and it now operates on finished goods rather than inputs. An April 2026 proclamation revised

Section 232 so derivative articles substantially made of steel, aluminum or copper face 25% on full value – the taxable base shifting from raw metal to the full value of the finished product, with the duty applying to goods containing 15% or more steel. Because only around a quarter of appliances sold in the US are domestically produced on an industry-average basis, exposure is near-universal, and tariffs reach the steel, compressors and control boards inside units assembled domestically as well. Component inflation compounds it: the memory shortage running through 2026 raises the bill of materials on precisely the connected products the industry is betting on. Asian manufacturing concentration, long the sector's cost advantage, is now its principal vulnerability – and the mitigation is expensive. Dual sourcing, North American localization and buffer inventory all lengthen the cash conversion cycle while financing costs stay elevated, and they concentrate obsolescence risk in a category where regulation mandates decade-long parts support.

A widening profitability gap amid competitors: quality premium is still resilient. Margin outcomes are diverging sharply, and the split is the most investable feature of the sector. Premium and mid-premium hold up because quality, reliability and design remain genuine purchase arguments and support pricing, and manufacturers reported early-2026 growth concentrated specifically in premium products, subscriptions and services, and B2B, with selective pricing. The volume tier is a different business: Chinese brands continue taking major and small appliance share in key Western European markets with high-functionality products at competitive prices, and as US demand weakens under rising prices, those manufacturers are explicitly redirecting effort toward Europe and other markets to offset it – meaning European price competition intensifies precisely as European volumes stagnate. Pass-through capacity exists but is thin and reputationally costly; where it has been used, it has come at the expense of units. Dividend policy and covenant headroom will be the leading distress indicators, and we expect consolidation across the mid-market where neither scale nor premium perception provides shelter.

Sustainability regulation: overcoming that hurdle would create new opportunities. The European framework is genuinely dual-edged and the second edge is underappreciated. All member states must transpose the Right to Repair Directive by 31 July 2026, and the obligation applies to products sold before that date – a retroactive service liability few have fully provisioned. But compliance costs raise entry barriers for import-dependent brands and favor regional manufacturers with embedded processes, while spare-parts obligations and mandated firmware updates make total cost of ownership the decisive purchase factor – shifting competition away from headline price, exactly where low-cost entrants are strongest. Layered onto already-high capital intensity and certification requirements, this is the most credible non-tariff defense Europe possesses.

It also creates the revenue stream the title points to: an installed base with legally guaranteed parts demand, consumer behavior aligned rather than resistant and service revenue delivering margin resilience even without unit growth – high-margin, capital-light and counter-cyclical, since it grows when hardware sales fall.

AI and connected products: the hoped-for renewal catalyst, still unproven. The industry's strategic bet is that embedded intelligence restarts the replacement cycle, and adoption is real – smart features now appear in a large majority of new launches and dominate product roadmaps. But the evidence for a pricing premium is weaker than the narrative: satisfaction research associates smart and connected appliances with more reported issues and user confusion. The credible value in the short run is more operational rather than promotional – remote diagnostics, predictive maintenance and subscription plans that raise customer lifetime value and feed the aftermarket annuity above.

Subsectors

Household Equipment comprises different industries involved in the design and manufacturing of domestic equipment for households. While companies exhibit very different profiles between segments, they have in common a strong sensitivity to consumer spending and a reliance on Asia-Pacific, mostly China, for their manufacturing activities. Mature economies still make up the vast majority of demand.

- **Consumer electronics:** Audio and video equipment, computers, mobile phones etc. Product markets are now largely mature and driven by replacement sales. A handful of companies generally dominates their markets and focus on design, R&D, marketing etc. while outsourcing manufacturing to contract manufacturers. Asia-Pacific concentrates over 80% of global turnover. Mexico is the only significant manufacturer outside of the region.
- **Household appliances:** Small appliances (home care, personal care, cooking aids) and large appliances (refrigerators, ovens, dishwashers, washing machines). Much like for consumer electronics, product markets are largely mature and dominated by a few large global players and smaller niche, regional competitors. Asia-Pacific's share of global turnover stands at about 80%. Other manufacturing countries include Germany, Italy, Turkey and the US.
- **Furniture and furnishings:** A highly fragmented industry dominated by small- and medium-sized companies working closely with designers and retailers. While Asia-Pacific still catches a substantial share of revenue (almost half), North America (18%) and few western European countries like Germany and Italy (20%) have a significant presence in the industry.

IT Services

How to adapt to AI-driven redistribution of value ?

Sector rating

L Low risk for enterprises

Strengths & weaknesses



- Low capex-intensive business
- Further digitalization/automation of process drives new demand for software
- External expertise claimed due to facilitate new technology integration/interoperability
- High-margin business due to compressible development costs/expertise premium
- No tariff barrier on services



- Low entry barrier / high competition
- Risks of substitution effects/revenue dilution due to broader adoption of GenAI agents for coding
- Rising AI inference costs cause risks of cost arbitrage on IT budget allocation
- Increasing obsolescence risks / Shorter lifespan of products
- Deteriorating investor sentiment over the industry = capital outflow

Sector overview

What to watch?

- Greater connectivity of the global economy is expected to drive new strong demand for software solutions and advisory services.
- Under further AI expansion, software companies will be forced to revamp their business model and shift from Software as a Service (SaaS) to Service as a Software model.
- An increase of specific or “industrial-focused” applications working under proprietary ecosystems under fears of low data protection from frontier AI models.
- Rising demand for cybersecurity solutions as corporates have low defensive lines against rising cyber threats caused by increasing LLM reasoning capabilities.
- Market structure revamping with higher share for hyperscalers and the emergence of new players surfing on easy-to-scale solutions working on open-weight infrastructure
- Investors’ sentiment toward software industry and resulting liquidity/solvency impacts.

Solid demand hiding critical changes in commercial mix.

Global demand looks supportive: digitalization, interoperability mandates and the migration of AI pilots into production keep spend flowing, but headline growth is modest and bifurcated. Global IT consulting and services grew roughly +3.6% in 2025, with a similar pace expected into 2026 – a low-single-digit market, not a boom. Within software, the split is sharp: smaller subscription cohorts are re-accelerating (the sub-USD50mn revenue tier bounced back to ~18% growth in 2025, near 2023 levels) while large-caps continue to moderate. Amid other early weak signals, retention business has drifted down to a median near 101%, versus ~105% earlier in the cycle, signaling that expansion within the installed base is thinning. The critical mix shift is commercial: the industry is being pushed from seat-based SaaS toward outcome- and consumption-priced “service-as-software,” where the vendor is paid for work delivered rather than licenses sold – a leading strategy house already books roughly a quarter of fees on outcome-based contracts. This decouples revenue from headcount and seat counts alike, rewarding those who can quantify delivered value and penalizing those still selling access.

The “AI deflation” vector is revamping market competition.

The large delivery-led services majors have themselves flagged that AI-native execution will dilute future revenue: rising adoption is intensifying pricing pressure, triggering deal renegotiations and slowing execution as clients reassess spend. Where value historically accrued to billable hours, agentic coding and automated delivery compress the fee pool directly. Compounding this, competitive walls are falling: hyperscalers are absorbing more of the platform layer, open-weight infrastructure lets new entrants scale agile and differentiated capital-light tools and frontier-model providers are moving directly into the advisory and delivery market. The defensible ground narrows to sector depth, proprietary data, regulated-industry mandates and integration complexity that a generic model cannot yet own. Companies with low compressible digital development and high staffing ratio work are the most exposed. The competitive question for 2026/2027 is less about who has the best AI than who can convert AI from a margin threat into a pricing-power moat before the fee pool reprices.

Profitability premium erosion under scrutiny. Core software still boasts exceptional gross margins – a median near 80% on software revenue (~76% blended) – and scaled players are extracting real operating leverage, with the USD500mn–USD1bn revenue cohort lifting EBITDA margins from ~17% to ~29% year-on-year through expense discipline. Services margins remain healthy but pressured, holding around 22–23% at the operating level, with providers warning that cushion narrows as talent costs rise and AI investment continues. The new drag is inference: AI-native products carry structurally lower gross margins because compute and inference costs are genuine, recurring COGS. The winners of 2026/2027 will be those who pass inference costs through pricing and cap usage, and improve unit economics as they scale – turning compute from a margin leak into a moat. The risk case is cost arbitrage on the client side: as inference becomes a visible IT budget line, CIOs may reallocate away from third-party software and services toward in-house or model-native alternatives, squeezing the sector’s addressable spend even where its own margins hold.

AI is dramatically shortening product lifespan; premium is stored in security and sovereign solutions. GenAI compresses the build cycle, which cuts both ways: faster time-to-market for incumbents, but also faster obsolescence and lower switching costs, eroding the recurring-revenue stickiness the whole valuation model rests on. Two structural demand pockets stand out against this. First, cybersecurity, where the threat surface is expanding mechanically with model capability – AI-enabled attacks have risen sharply (on the order of +70% year-on-year by some breach-tracking measures) – and defensive spend is one of the few line items still growing while budgets consolidate elsewhere: enterprise security budgets rose ~5% into 2026, with AI-specific security now exceeding 11% of the total. Second, data-sovereignty and industrial-specific applications: distrust of frontier models’ data handling is pushing regulated and industrial

clients toward proprietary, on-premise or verticalized ecosystems – a demand stream that favors providers with domain depth and controlled deployment. Both pockets reward specialization over horizontal reach and are the likeliest sources of durable, defensible growth over the outlook window.

Investors need to be reconquered. Investor sentiment has deteriorated materially in 2026. Leading pure-play names fell on the order of 25–30% even as they reported solid results, a sell-off driven not by earnings but by a collapse in confidence in the business model itself. Under a “SaaS-Apocalypse” mood, public investors concluded they lacked clarity on how to price AI risk – the fear that frontier-model capabilities and falling entry barriers erode software’s growth, stickiness and pricing power. Valuations de-rated to distressed levels: a bellwether integrator shed ~40% of its value across 2025 and now trades below the broader IT-sector earnings multiple (roughly 17x), stripped of its long-standing quality premium. Such sentiment deterioration and flight of capital out of software industry matters because capital is the sector’s lifeblood for building out structure and funding future growth: a higher cost of equity, colder IPO and M&A windows and cautious sponsors hit the sub-scale, growth-funded tier hardest, where cheap capital was the model’s oxygen. Crucially, solvency is not the concern – sentiment is. Asset-light economics, high free-cash-flow conversion and predominantly net-cash balance sheets let profitable incumbents absorb a valuation reset without liquidity stress. Expect a “rich-get-richer” bifurcation: cash-generative leaders consolidating and self-funding their next phase, while capital-starved, weak-retention challengers face an existential funding squeeze.

Subsectors

The industry can be divided into five main subsegments - companies generally operate across many segments. While IT consulting and programming is a fragmented activity, the managed services, software and data processing segments are far more concentrated with regional and global players.

- **Consulting** is about helping organisations make the best possible use of digital technologies to operate more efficiently.
- **Programming** is about developing custom digital tools to help organisations reach their business objectives.
- **Managed services** is about outsourcing the management of the IT infrastructure to a third-party company.
- **Software** are complex, generic programmes that can be made available to a variety of corporate customers.
- **Data processing**, including cloud computing, is about collecting and treating large quantities of data for corporate customers.

Machinery and Equipment

Approaching a cyclical inflection point

Sector rating

M Medium risk for enterprises

Strengths & weaknesses



- High barriers to entry due to significant capital investment, engineering expertise, proprietary technology and extensive service networks.
- Robust long-term growth potential, given structural trends such as industrial automation, robotics, artificial intelligence (AI), electrification and digital manufacturing
- Pricing power through technological differentiation allows resilient margins.
- Diversified revenue streams reduces dependence on any single customer segment or economic cycle.
- Strategic role in industrial digitalization through connected equipment, industrial software, data analytics and AI-enabled solutions.
- Growing recurring revenue from after-sales services which enhance customer retention and support margin resilience.



- High cyclical exposure: Demand is closely tied to industrial activity, capital expenditure and business confidence.
- Complex global supply chains increase exposure to geopolitical risks, trade restrictions and component shortages.
- High capital intensity required to maintain competitiveness.
- Raw material price volatility, particularly in key industrial inputs, including steel, aluminum, copper and nickel.
- Long sales and delivery cycles can delay revenue generation and reduce operational flexibility.
- Stringent environmental, safety and cybersecurity requirements increase compliance costs and complexity.
- Rapid technological change requires continuous innovation, while legacy assets risk becoming obsolete.

Sector overview

What to watch?

- **Global economic outlook:** Macroeconomic conditions continue to weaken. Following modest growth of just +2.9% in 2023 and +2.8% in 2024, global GDP growth is projected to remain sluggish in 2025 and 2026. This prolonged slow recovery could constrain sector growth in the near term, particularly given its sensitivity to economic cycles.
- **Business confidence & manufacturing activity:** Key indicators such as business confidence and Purchasing Managers' Indexes (PMIs) serve as vital barometers for sector health. Over 2024 and the first half of 2025, these indicators have showed a relatively flat evolution, with no clear sign of improvement.
- **Evolving industrial policies:** Major economies like the US, China, India, Germany and Brazil are ramping up industrial policies and subsidies. Governments are increasingly shaping industrial priorities and strategically supporting innovation and technology adoption, which could reshape competitive dynamics and investment flows within the sector.
- **Supply-chain normalization:** The severe supply-chain disruptions of 2020-2022 have notoriously eased throughout 2023 and 2024, aided by a slowdown in global trade volumes. This relief has been particularly beneficial for machinery manufacturers, who depend heavily on a global network of parts suppliers.

- **Commodity price volatility:** Metal prices remain volatile. While steel and nickel prices have softened, aluminum and copper prices continue showing upticks. These fluctuations present ongoing margin risks for manufacturers reliant on these raw materials.
- **Disinflation and better financial conditions:** Inflationary pressures have eased, with global inflation expected to moderate. Thanks to this, financing conditions should continue easing throughout the year in Europe particularly. However, companies in the US should remain cautious as the trade war could increase prices in the country.
- **Accelerating AI adoption:** Leading firms are rapidly integrating AI and machine learning into production, driving efficiencies and enabling the development of advanced machinery. The ongoing wave of industrial automation presents substantial growth opportunities and productivity gains.
- **Sustainability and regulatory pressures:** Increasing global focus on environmental regulations and sustainable manufacturing practices is driving demand for greener machinery and energy-efficient equipment. Compliance costs and the pace of adopting sustainable technologies will be crucial factors to monitor.
- **Geopolitical tensions and trade policies:** Ongoing geopolitical uncertainties and evolving trade policies may disrupt supply chains and market access. Machinery producers need to stay agile in navigating higher tariffs, export controls and shifting alliances.

The global machinery & equipment sector is valued at approximately USD820bn in 2026. Asia-Pacific remains the largest regional market, supported by China and India's industrial expansion, while North America and Europe continue to lead in technological innovation, high-value equipment, and installed industrial capacity.

As of 2026, the sector is entering the early stages of a recovery following a prolonged period of cyclical weakness. Following two years of subdued order activity, the global machinery & equipment sector is showing early signs of recovery. During 2024 and 2025, new orders weakened as higher interest rates, softer industrial activity and geopolitical uncertainty delayed capital expenditure decisions. Manufacturers were able to maintain production by working through elevated backlogs accumulated during the post-pandemic recovery (2021–2022), leading to a gradual normalization of order books. As inventories rebalance and investment conditions improve, demand is expected to gradually recover in 2027, supported by infrastructure spending, automation, AI-related investment and industrial digitalization.

Indeed, throughout the first half of 2026, order intake has started to improve in some sub-sectors, given increasing investment in power infrastructure, data centers, industrial automation and reshoring initiatives. Although demand remains uneven across end markets, the sector appears to have reached a cyclical inflection point.

Production trends reflect a similar pattern. Following the strong post-pandemic rebound, global machinery output moderated significantly during 2024 and flattened in 2025 as customers delayed investment decisions and manufacturers reduced elevated inventories. Entering the second half of 2026, production is gradually recovering as capacity utilization improves and replacement demand strengthens. Construction machinery and heavy-duty trucks are expected to lead the recovery, benefiting from infrastructure spending, expanding power generation and data center investments, healthier freight markets and improving dealer inventories. By contrast, agricultural machinery is likely to remain the weakest major end market in the near term, with subdued farm economics and export uncertainty continuing to constrain equipment investment despite signs that the cycle is approaching its trough.

Regionally, North America continues to outperform most other markets, supported by infrastructure investment, manufacturing reshoring and robust demand for power generation and digital infrastructure. Europe should be timidly recovering after a prolonged period of weak industrial activity, helped by easing inflation, lower financing costs and improving business confidence, although activity remains below the post-pandemic highs. Asia-Pacific continues to benefit from long-term industrialization, automation and infrastructure investment, with China and India remaining key drivers of equipment demand.

Although geographically uneven, the global macroeconomic environment should also become increasingly supportive for the sector. Global GDP is projected to expand by +2.9% in 2027 (after a temporary slowdown in 2026), providing a more favorable backdrop for industrial investment and machinery demand. Regional dynamics remain mixed: The US economy is expected to remain resilient, growing by around +2.0–2.1% in 2026, while Europe faces the slowest recovery (+0.9% in 2026; +1.2% in 2027). China remains

Subsectors

The machinery & equipment sector includes the following sub-sectors: construction machinery, agricultural & farm machinery, industrial machinery and equipment, mining equipment and robotics.

Metals

Capital discipline sustains balance sheets as energy transition creates long-run demand

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Long-run demand for critical metals – lithium, cobalt, nickel, copper, rare earths – is underpinned by structural growth in EV manufacturing, renewable energy infrastructure and semiconductor production, with demand for critical minerals projected to triple by 2030 under net-zero scenarios.
- Many major mining companies emerged from the 2021 commodity boom with strong balance sheets, having accumulated cash and reduced debt rather than pursuing aggressive expansion; this financial resilience provides a buffer against market volatility.
- Governments globally are prioritizing critical mineral security, creating policy tailwinds in the form of expedited permitting, strategic reserve purchases and direct investment support for domestic production.
- Base metal prices, though off their 2022 peaks, remain above long-run historical averages for many categories, sustaining industry profitability at current production levels.



- The sector's supply chains are highly globalized and therefore vulnerable to geopolitical shocks: trade disputes can rapidly translate into export restrictions on critical inputs, as demonstrated by China's curbs on rare earths and graphite exports.
- Escalating ESG and environmental regulatory requirements – covering carbon emissions, land use, water management and community relations – are increasing compliance costs and contributing to project delays, particularly for new mine development.
- The sector is inherently capital-intensive with long project lead times (over 10 years from discovery to production for new mines), making it difficult to respond quickly to demand signals and creating structural supply-demand mismatches.
- Commodity price volatility remains high: base metal and bulk material prices are closely tied to Chinese industrial demand, global growth cycles and financial investor positioning, all of which can shift rapidly.

Sector overview

What to watch?

- Critical mineral policy and resource nationalism: Governments are treating strategic minerals as national security assets, leading to new export controls, domestic content requirements and investment screening measures that are reshaping global supply chains.
- China's industrial demand trajectory: China consumes over two-thirds of the world's base metals and iron ore. Its construction and manufacturing activity – and the pace of its own domestic decarbonization – will continue to set global metal price trends.
- Financing conditions and capex discipline: After the 2021 boom, the sector has maintained conservative capital allocation. Any shift toward aggressive expansion would represent an early signal of supply growth that could eventually weigh on prices.
- Energy transition metal demand: The ramp-up of EV production, battery storage, wind turbine installation and grid expansion is creating accelerating demand for copper, lithium, nickel and cobalt that differs structurally from traditional industrial metal demand.
- Environmental regulation and permitting: Stricter environmental standards and community opposition are making it increasingly difficult and time-consuming to bring new mines into production in key jurisdictions, constraining supply response even when prices are supportive.

The global metals and mining sector supplies the essential raw materials – from iron ore and steel to copper, gold, lithium and rare earth elements – that underpin industrial development and the global energy transition. In 2025, gross output for the basic metals and metal products industry reached approximately USD9.6trn globally. Asia-Pacific, dominated by China, accounts for around 60% of the global market. Europe is a distant second at approximately 21%. This regional concentration reflects China's position as by far the largest producer and consumer across the metals spectrum, from steel and aluminum to rare earths. Gross output for basic metals and metal products is estimated at approximately USD10.4trn in 2026, with the 2026-2028 gross CAGR projected at approximately +7%.

After windfall profits during the 2021 commodity super cycle, the sector experienced a normalization in financial performance through 2023-2024. Prices for many base metals and bulk materials retreated from their peaks, while input costs (energy, reagents, labor) remained elevated. The aggregate EBITDA margin for the global metals and mining sector sat at approximately 15% in 2025, reflecting continued normalization from the elevated levels of the 2021 boom. Companies have responded with capital discipline: excess cash from the boom was largely returned to shareholders rather than committed to new capacity, keeping balance sheets strong but constraining future supply growth. This conservative posture is maintaining profitability at current price levels but may create a supply shortfall as energy transition demand for critical minerals accelerates.

The structural long-term demand story for the sector is compelling, particularly for the subset of metals critical to the energy transition. Copper, which is essential for electrical wiring in EVs, wind turbines and grid upgrades, is expected to face significant supply deficits as demand accelerates faster than new mine capacity can be developed. Lithium demand is projected to grow eight-fold by 2040 under net-zero scenarios. Rare earth elements are increasingly strategically contested, with China's dominance in processing creating a

geopolitical dimension that is driving significant investment in alternative supply chains in the US, Europe and Australia. For bulk materials – iron ore and steel – the picture is more mixed, with Chinese demand remaining the swing factor and the energy transition simultaneously creating both headwinds (reduced steel demand as fossil fuel infrastructure investment declines) and tailwinds (new steel demand for wind towers, EV infrastructure and buildings retrofit).

Subsectors

Steel production: The world's most widely used metal, critical for construction, automotive and industrial applications. The largest single segment of the metals sector by volume; increasingly subject to decarbonization pressure as steelmaking accounts for approximately 7% of global CO₂ emissions.

Base metal mining: Mining of non-ferrous metals including copper, aluminum, zinc, nickel and lead. The base metal mining market was valued at approximately USD550bn in 2024 and is projected to reach USD750bn by 2030, driven by energy transition demand.

Iron ore mining: Mining of iron ore for steelmaking. Demand is closely correlated with Chinese steel production and global construction activity.

Metal processing and manufacturing: Processing of raw metals into finished products and components, including aluminum rolling, copper wire drawing and precision metal parts. Closer to end markets than mining and more sensitive to manufacturing cycle fluctuations.

Precious metals mining: Mining of gold, silver and platinum group metals, primarily for jewelry, investment and industrial applications. Gold has traded near all-time highs as investors seek inflation hedges and safe-haven assets.



Paper

Stabilization underway

Sector rating

 Sensitive risk for enterprises

Strengths & weaknesses



- E-commerce growth continues to support demand for paper-based packaging, particularly corrugated boxes and other fiber-based solutions, driven by the expansion of online retail and logistics activity.
- Global population growth and rising living standards support long-term demand for essential paper products, especially household paper goods such as tissue products and hygiene-related papers.
- Growing consumer and regulatory pressure to reduce plastic waste is accelerating the adoption of recyclable and biodegradable paper-based packaging solutions, creating new growth opportunities for fiber-based alternatives.
- A well-established recycling ecosystem enhances the sector's circular economy profile, improving resource efficiency, reducing dependence on virgin fibers and lowering raw material costs.
- A diversified product portfolio across packaging, tissue, printing and specialty papers provides market resilience by reducing exposure to downturns in individual segments.



- Digitalization continues to structurally reduce demand for traditional graphic paper products, including printing, publishing and office papers.
- Environmental concerns related to forestry practices, emissions and resource consumption create reputational and regulatory risks for industry participants.
- The sector's high energy intensity makes profitability vulnerable to fluctuations in energy and chemical input costs, particularly in regions with higher operating expenses.
- Increasingly stringent environmental regulations and sustainability requirements require continuous capital investment in cleaner technologies, emissions reduction and process improvements.
- Dependence on sustainable fiber supply exposes companies to supply-chain risks, including disruptions caused by geopolitical tensions, climate change, forest management challenges and fluctuations in recycled fiber availability.

Sector overview

What to watch?

- Evolution of e-commerce: Online retail is expected to remain a key driver of demand for paper-based packaging, particularly corrugated boxes and containerboard. However, the pace of e-commerce growth, changing logistics models and packaging optimization efforts will influence future demand.
- Packaging regulation and plastic substitution: Governments – particularly in Europe, but increasingly in other regions – are introducing regulations aimed at reducing single-use plastics and promoting recyclable and fibre-based packaging. The pace and scope of these policies will shape product innovation, investment priorities and market growth.

- **Energy prices and competitiveness:** Although energy markets have stabilized since the 2022–2023 crisis, electricity and natural gas prices remain elevated in several regions, particularly Europe as an aftermath of the 2026 conflict in the Middle East. Future energy price trends and decarbonization policies will continue to affect production costs, investment decisions and the global competitiveness of manufacturers.
- **Demand for tissue and hygiene products:** Rising populations, urbanization and increasing hygiene awareness – particularly in emerging economies – are expected to support long-term growth in tissue and other household paper products. Demographic trends and public health policies will remain important demand drivers.
- **Digital transformation:** The continued shift toward digital media, electronic documentation and remote working is expected to further reduce demand for graphic papers. The speed of this transition will determine how quickly producers need to rebalance their product portfolios toward packaging and specialty papers.
- **Construction and industrial activity:** Trends in residential construction, renovation and broader industrial production – particularly in major markets such as the US, China and Europe – will influence demand for wood products.

The global pulp and paper industry remains an important component of the global economy, supplying essential materials across packaging, hygiene, industrial and communication applications. The sector is characterized by a diversified product portfolio, ranging from fiber-based packaging and tissue products to pulp, specialty papers and graphic papers. From a geographical perspective, the sector is concentrated across Asia-Pacific, North America, Northern Europe and Latin America, each with distinct competitive advantages. APAC, led by China, represents the largest production and consumption hub, supported by its extensive manufacturing base, strong packaging demand and large domestic market. North America, particularly the US and Canada, remains a key producer across pulp, packaging, tissue and wood products, benefiting from abundant forest resources and integrated industrial assets. Northern Europe, led by Sweden and Finland, maintains a strong position in sustainable forestry, market pulp and specialty paper production, although higher energy and wood costs have challenged competitiveness. Meanwhile, Latin America, especially Brazil and Chile, has become a global powerhouse in pulp production, supported by large-scale plantations,

modern mills, and some of the lowest production costs worldwide.

While the industry continues to face structural challenges, particularly from digitalization and changing consumption patterns, it is undergoing a significant transformation driven by sustainability trends, packaging substitution, recycling expansion and operational optimization.

Packaging remains the largest segment of the industry, benefiting from the continued growth of e-commerce, consumer goods demand and the substitution of plastic-based solutions with recyclable fiber alternatives. Tissue and hygiene products continue to demonstrate defensive characteristics, supported by population growth, urbanization and rising hygiene standards, particularly in emerging markets. Conversely, graphic papers continue to experience structural decline as digital communication and reduced office printing demand accelerate the shift away from traditional paper grades.

The industry outlook has improved compared with the challenging conditions observed during 2022–2024.

However, the recovery remains uneven and is primarily supported by supply-side actions rather than a strong acceleration in demand. Following a period of oversupply, weak pricing and inventory adjustments, producers have implemented capacity reductions, mill closures, cost-cutting programs, and operational efficiencies to restore market balance. As a result, profitability is expected to gradually improve in 2026–2017, although demand growth remains modest across most segments.

The pulp market is showing signs of recovery after a period of significant price pressure caused by excess capacity and weaker demand, particularly in China. Supply discipline, production curtailments and limited near-term capacity additions are expected to support a gradual improvement in pulp prices. Latin American producers remain structurally advantaged due to lower production costs and newer assets, while European producers continue to face higher wood and energy costs.

Packaging markets are expected to remain relatively resilient, although growth is likely to be moderate as the global economy sees weak growth. Industrial packaging demand is forecast to remain generally stable, supported by e-commerce trends, while consumer packaging continues to face pressure from weak consumer confidence and excess capacity in certain grades. The sector's long-term growth opportunity remains linked to the substitution of plastics and the increasing adoption of sustainable packaging solutions. The tissue segment continues to provide stability to the industry due to its essential nature and lower cyclicity.

Demand is expected to grow broadly in line with population trends in developed markets and at a faster pace in emerging economies as hygiene standards improve.

Nevertheless, profitability remains exposed to fluctuations in pulp, energy and transportation costs. In contrast, graphic paper (print and newspaper) remains and will continue to be the most challenged segment. The structural decline caused by digitalization continues to weigh on volumes, despite capacity rationalization and selective pricing improvements. Producers are increasingly converting graphic paper capacity toward more attractive segments, particularly packaging and specialty papers, to adapt their portfolios to changing market conditions.

Financial performance has improved from the downturn experienced in the previous years, but the sector remains below historical profitability levels.

The earnings recovery is expected to be gradual, supported by lower-cost structures, capacity discipline and improved pricing in selected segments rather than by strong market growth. The prolonged geopolitical uncertainty, including the ongoing conflict in the Middle East, represents an additional risk factor by increasing exposure to higher energy, freight, chemical and logistics costs. Under a prolonged escalation scenario, weaker global growth and renewed inflationary pressures could delay the sector's recovery. Overall, the global pulp and paper industry is transitioning from a period of contraction and restructuring toward a more stable operating environment. Long-term opportunities remain concentrated in sustainable packaging, tissue products, specialty papers and circular economy solutions. However, companies' ability to adapt through portfolio diversification, cost competitiveness, energy efficiency and sustainable innovation will remain critical in determining long-term competitiveness.

In terms of financial performance, the past two years were tough for the sector. Revenues fell by around -14% y/y on average in 2023 and timidly recovered by +4% y/y in 2024. The sector had a five-year historical compound annual growth rate of sales of +6%. This drop was driven by a contraction of both prices and volumes due to the weakening of the global economy. Indeed, paper (together with other commodities) lost a lot of pricing power last year amid increasing economic uncertainties. In parallel, the pulp & paper sector is energy- and raw-materials-intensive, notably dependent on chemicals and natural gas for transforming raw wood into pulp and then into other derived sub-products. Therefore, companies have seen margins squeeze in the past years. As a fact, operating margins moved from a five-year historical average of 8.8% to only 6.2% in 2024. The sector's margin deterioration was particularly exacerbated in Europe, where energy prices increased the most. Nevertheless,

operating conditions started to improve gradually since Q4 2024 and throughout 2025, indicating that the worst is already behind, even if geopolitical tensions persist. For 2025, we expect revenues to recover by around +4.3% y/y, supported by the fact that destocking in many end-markets has ended and the belief that consumer e-spending should gradually start to improve as inflation has receded.

Subsectors

Paper packaging: This includes containerboard, corrugated packaging, cartonboard, kraft paper and other fiber-based packaging products used across e-commerce, food and beverage, consumer goods and industrial applications. Despite softer conditions across most paper markets, packaging remains the industry's most resilient segment, supported by steady demand growth.

Graphic papers (printing, writing & newsprint): Comprises coated and uncoated printing and writing papers, newsprint and other graphic paper grades used for publishing, office and educational applications. Graphic papers continue to experience structural decline as digitalization reduces demand for printed media and office paper.

Pulp: Produces virgin pulp, the primary raw material used in paper, packaging, tissue and specialty paper manufacturing. The pulp market is capital-intensive and globally traded, with profitability closely linked to supply-demand dynamics and commodity price cycles.

Tissue & hygiene paper: Includes household and away-from-home tissue products such as toilet paper, paper towels, facial tissues and napkins. Tissue is the industry's most defensive segment, benefiting from stable, non-discretionary demand.

Forest products (wood & lumber/timber): Includes timberland management, lumber, engineered wood products, plywood, oriented strand board and other wood-based materials primarily used in residential construction, infrastructure and industrial manufacturing. This segment is highly cyclical, with demand closely linked to housing starts, renovation activity and broader construction markets.

Pharmaceuticals

Navigating innovation, pricing pressures and an evolving geopolitical landscape

Sector rating

L Low risk for enterprises

Strengths & weaknesses



- Global, diversified and resilient market, supported by structural non-cyclical demand for healthcare.
- High barriers to entry due to significant R&D investment, stringent regulatory requirements, intellectual property protection and specialized expertise limit.
- Strong profitability and cash generation, supporting continued innovation, acquisitions and shareholder returns.
- Favorable long-term demand drivers, including aging populations, increasing prevalence of chronic diseases and expanding healthcare access in emerging markets.
- Pricing power through innovation, partially offsetting inflationary and cost pressures.
- Innovation leadership strengthen long-term growth potential.



- High regulatory burden increases costs, extends development timelines and adds operational complexity.
- High R&D risk: Drug development requires substantial investment with long development cycles and high clinical failure rates.
- Patent dependency, increasing reliance on a strong and continuous innovation pipeline.
- Governments and healthcare systems are increasingly focused on controlling drug costs, limiting pricing flexibility and pressuring margins.
- Competition from generics and biosimilars.
- Supply-chain complexity increases exposure to geopolitical tensions, trade restrictions, and supply disruptions.
- Reputational and ESG risks.

Sector overview

What to watch?

- Aging populations and rising chronic diseases: Demographic trends remain one of the strongest long-term growth drivers for the pharmaceutical industry. A rising population (8.5bn people by 2030), together with increasing life expectancy and a higher prevalence of chronic conditions such as cardiovascular disease, diabetes, cancer and neurodegenerative disorders, is expected to sustain demand for innovative therapies over the coming decades.
- GLP-1 therapies reshaping the market: Anti-obesity and diabetes drugs continue to transform the industry. Beyond treating obesity and type 2 diabetes, ongoing clinical trials are exploring their potential in cardiovascular disease, chronic kidney disease and other indications, significantly expanding their addressable market and creating one of the fastest-growing therapeutic segments.
- Artificial Intelligence across the pharmaceutical value chain: AI is rapidly transforming drug discovery, clinical trial design, regulatory processes and commercial operations. By improving target identification, optimizing clinical trials and accelerating data analysis, AI has the potential to reduce development timelines, improve R&D productivity and lower costs.
- Patent cliffs and pipeline execution: A growing number of blockbuster drugs are approaching patent expiration over the next few years, exposing pharmaceutical companies to increased generic and biosimilar competition. Sustaining growth will depend on the successful commercialization of innovative products, strategic acquisitions and strong late-stage development pipelines.

- **Expansion of biologics and precision medicine:** Biologics, cell and gene therapies, RNA-based medicines and personalized treatments continue to reshape healthcare by targeting diseases with greater precision. At the same time, increasing biosimilar adoption will intensify competition while expanding patient access to advanced therapies.
- **Regulatory evolution and faster access to innovation:** Regulators continue to modernize approval pathways for breakthrough therapies, rare diseases and unmet medical needs. While regulatory requirements remain rigorous, initiatives aimed at accelerating reviews and incorporating real-world evidence could shorten time-to-market for innovative medicines.
- **Supply-chain resilience and manufacturing diversification:** Pharmaceutical companies are continuing to diversify manufacturing footprints and strengthen supply chains in response to geopolitical tensions, trade restrictions and lessons learned from recent disruptions. Securing critical raw materials and active pharmaceutical ingredients (APIs) has become an increasingly strategic priority.
- **Pricing and healthcare affordability:** Governments and payers are placing greater emphasis on controlling healthcare expenditure through pricing negotiations, reimbursement reforms and increased use of generics and biosimilars. Companies will need to balance innovation with affordability while demonstrating the clinical and economic value of new therapies.

The pharmaceutical industry remains one of the most resilient and innovation-driven sectors globally, supported by structural demand stemming from ageing populations, rising prevalence of chronic diseases, expanding healthcare access and continuous scientific advances. While the sector continues to be dominated by large multinational companies headquartered in the US and Europe, innovation is increasingly supported by biotechnology firms, strategic partnerships and targeted acquisitions, reinforcing a highly dynamic research ecosystem.

Following several years of exceptional innovation, the industry has entered a new growth phase driven less by pandemic-related products and increasingly by breakthrough therapies addressing large unmet medical needs. Metabolic diseases, oncology, immunology, rare diseases and precision medicine remain the main pillars of pharmaceutical innovation, while artificial intelligence is progressively improving productivity across the entire

value chain – from drug discovery and clinical trial design to manufacturing and commercialization. These advances are expected to improve R&D efficiency and support a stronger pipeline of high-value therapies over the medium term.

One of the industry's strongest growth engines continues to be obesity treatments. The rapid adoption of next-generation metabolic therapies has created one of the fastest-growing pharmaceutical markets in history, with global sales expected to multiply over the remainder of the decade as reimbursement expands, oral formulations become commercially available and patient access improves across developed markets. At the same time, continued investment in biologics, cell and gene therapies and RNA-based medicines is broadening treatment options for previously difficult-to-treat diseases, supporting long-term sector growth despite increasing biosimilar competition.

From a macroeconomic perspective, the sector is navigating a more complex operating environment. Geopolitical tensions and evolving trade policies are encouraging pharmaceutical companies to diversify manufacturing capacity and strengthen supply-chain resilience, particularly for active pharmaceutical ingredients and strategically important medicines. At the same time, discussions around tariffs, domestic manufacturing incentives and government procurement policies are prompting companies to reassess production footprints and capital expenditure plans, particularly in the US. Although the ultimate scope of these measures remains uncertain, they reinforce the ongoing trend toward greater regionalization of pharmaceutical manufacturing.

Drug pricing will remain one of the key issues shaping industry profitability. Governments and healthcare systems continue to seek greater affordability through price negotiations, reimbursement reforms and wider adoption of generics and biosimilars. Nevertheless, recent policy developments suggest that pricing pressure may prove less severe than initially anticipated, while broader reimbursement for innovative therapies is expected to support higher treatment volumes. Consequently, the industry's revenue outlook increasingly depends on expanding patient access and launching differentiated medicines rather than relying solely on price increases.

Patent expirations will continue to represent a structural challenge over the coming years, although the financial impact may be more gradual than previously expected. The increasing complexity of biologics, advanced formulations and drug-delivery technologies has made generic substitution more difficult and, in many cases, has effectively extended the commercial life of leading products. As a result, pipeline quality, regulatory execution and successful product launches are expected to become more important drivers of earnings than patent expirations alone.

Looking ahead, the sector's outlook for 2027 and beyond remains constructive. Demand fundamentals continue to strengthen, supported by favorable demographics, rising healthcare expenditure and sustained innovation. Artificial intelligence is expected to improve research productivity, while strategic acquisitions will likely remain an important mechanism for replenishing pipelines and accessing emerging technologies. Although geopolitical uncertainty, evolving regulation and pricing reforms will continue to generate volatility, the pharmaceutical industry appears well positioned to deliver resilient earnings growth over the medium term, underpinned by strong cash generation, high barriers to entry and one of the most attractive innovation pipelines across the global economy.

Subsectors

Drugs manufacturers:

- a. **Active Pharmaceutical Ingredients (API):** API is the part of any medication that produces the intended health effects. They are produced from raw materials with a specified strength and chemical concentration. Examples of APIs include: ibuprofen, loratadine, omeprazole and acetaminophen. All drugs are made up of two core components (APIs and excipients). The excipients are chemically inactive substances, such as lactose or mineral oil in the pill, which are used to help the medication remain stable and to control its absorption.

- b. **Patented drugs (also known as branded or original drugs):** Original drugs refers to drugs that have been approved for marketing after many tests and rigorous clinical trials (there are four phases before the post-market monitoring). This takes about 15 years of R&D and millions of dollars of investments. Only large multinational pharmaceutical firms are able to develop original drugs, which benefit from a patent period of 20 years. But once the patent expires, other manufacturers can start to produce generic versions of the drug.
- c. **Generic drugs:** Drugs that are not branded but that are very similar to a branded drug in terms of dosage, administration, safety and performance. Generic drugs tend to be cheaper and therefore more accessible, since their manufacturers did not have to invest in discovering/creating a new formula but only replicate an existing one.
- d. **Biotechnology:** This is the merging of biology and technology, referring to the branch of applied science that uses living organisms and molecular biology to produce healthcare-related products. Today, approximately 17% of total drug revenue is derived from biopharmaceuticals, which are mostly used in oncology, metabolic disorders and infectious diseases. The term biopharma describes companies that use both biotechnology and chemicals in their R&D.

Contract Research and Manufacturing Services

(CRAMS): The field encompasses those services in the pharmaceutical and biotechnology industries that require extensive R&D and large-scale manufacturing facilities. It is a clinical term used for referring to outsourcing and it is one of the fastest-growing segments in the sector today.

Drug marketing: This refers to the marketing of drugs and medical devices by private and public organizations to doctors, clinicians and consumers as drugs/treatments need to adopt particular marketing strategies to be sold effectively.



Retail

Monetizing data and shopper attention amid heightening pressure on consumption

Sector rating

 Sensitive risk for enterprises

Strengths & weaknesses



- High savings rate in Europe as consumption buffer
- State support for capping energy prices and protecting household purchasing power
- Stronger digital footprint on retail sales (revenue diversification + logistics savings)
- Gradual infusion of AI tools to boost revenue and optimize costs
- Increasing data collection improving product mix, shortening inventory turnover and strengthening retention



- Harsh competition from large marketplaces
- Supply-chain vulnerability under new tariff barriers
- Higher import cost amid higher energy prices and production bottlenecks on Asian supply chains
- New inflationary pressure and higher interest rates dampening consumer sentiment
- Weaker brand awareness (notably from younger generations)

Sector overview

What to watch?

- Consumer savings/purchasing attitude toward new inflationary pressure
- Inventory risk management along geopolitical tensions (consumer electronics first in line of impact from Middle East tensions within the retail segment)
- Evolution of trade tariffs and local rules on sourcing conditions/requirements
- Inclusion of gen-AI agents and tools to increase retail traffic and digital referencing, improving forecasts and optimizing inventory costs
- Growth development of second-hand market and positioning on traditional retailers in that segment
- Development of new customized services along customer journey to improve loyalty and brand awareness
- Customer preference for services over durable goods amid deteriorating geopolitical/political climate and purchasing power squeeze

A value-seeking consumer, spending sideways. The defining feature of 2026/2027 is not weak demand but demand that has changed shape. US retail sales are running above their pre-pandemic decade average, suggesting a stronger-than-normal year supported by income growth, employment stability and first-half fiscal relief. Yet growth is concentrated in higher-income households, leaving the revenue base exposed to any reversal in equity-driven wealth effects while lower-income cohorts trade down. Europe is slower but structurally sound, with wholesale and retail trade among the stronger-growing value-added sectors as the recovery broadens from services back into goods. However, in Europe households still save well above pre-pandemic norms (14% of disposable income vs. 12.5% before Covid-19). Such a high savings rate is a latent buffer but not a present strength as it is deferred consumption that has repeatedly failed to convert so it will not release on better sentiment alone. Energy-price intervention protects the essentials basket, not the discretionary categories where margin actually lives. Add

the drift from durable goods toward services and experiences and the result is nominal resilience masking gross-profit erosion: around four-fifths of consumers globally report trading down into private label, resale and off-price.

Trade fragmentation and the new inventory doctrine. Merchandise economics were already fragile well ahead of tariffs as grocery and general merchandise retail historically run at low-single-digit margins. They compressed further near breakeven levels under tariff-inflated landed costs, wage floors and persistent discounting. Against this new background, most supply-chain leaders have shifted sourcing toward tariff-neutral countries and a large majority of them are building buffer inventory, with onshoring and supplier diversification now mainstream. This is rational but expensive in three ways at once: 1) it lengthens the cash conversion cycle while financing costs stay elevated, 2) it concentrates markdown and obsolescence risk where gross margin is thinnest (ex: consumer electronics retail due to long lead times, concentrated sourcing and rapid ageing) and 3) it demands capital and vendor reach options mid-sized operators do not always have. Tightening customs treatment of low-value parcels, traceability requirements and European circularity rules raise compliance costs while removing sourcing flexibility. As a result, liquidity, not demand, is the binding constraint for the weaker half of the sector, and trade friction now functions as a share-transfer mechanism toward large balance sheets. Watch inventory days, markdown cadence and supplier terms to track solvency and liquidity risks on retailers, but also their suppliers.

Customer customized services and data monetization as a new high growth potential profit pool. The structural answer to thin merchandise margins is monetizing traffic rather than goods. Retail media is now a top-tier advertising channel with categorically different economics: advertising margins can exceed 50% against typical retail product margins of 2–5%, on revenue that is largely incremental because it monetizes existing traffic rather than cannibalizing product sales. Executives now treat these networks as crucial to both revenue and profitability, with expansion into non-endemic advertising underway. It is also capital-light, lifting ROIC and cash conversion without proportional investment – which is precisely what a low-margin, high-volume model needs. The same data asset underwrites the loyalty agenda: personalized services along the customer journey are the defensible response to brand versatility among younger cohorts, but profits become really tangible only where data scale already exists.

AI moving from pilot to operating system. Regular AI use is now near-universal among retailers, with a majority experimenting with agents, and the value is landing in working capital before revenue – better forecasting, fewer stockouts and markdowns, faster inventory turns, which

directly offsets the cash cost of the buffer-stock doctrine above. The strategic edge is agentic commerce: retailers expect AI agents to collapse discovery, decision and checkout into a single interaction, which may weaken brand loyalty and puts a premium on clean, machine-readable product and pricing data. Directionally, this is simultaneously a traffic opportunity for the digitally disciplined and an erosion of the browsing step where brand preference is formed.

Circularity as a margin and relevance play. Recommerce is compounding at high-single to low-double-digit rates, several times faster than traditional retail, explicitly driven by inflation pressure and weak sentiment, with Europe growing fastest on the back of circular-economy regulation. For incumbents, branded buy-back and resale keeps customers inside the ecosystem instead of ceding them to third-party marketplaces, earns margin on inventory already paid for, and pre-empts European repair and reuse mandates. The constraint is operational – sourcing consistency, authentication and reverse logistics are harder than forward retail – so treat it as a loyalty and relevance investment that turns margin-accretive at scale, not an immediate profit line.

Subsectors

Fast-moving consumer goods (food, personal care, house care, etc.): Traditionally the most resilient segment of the industry with little volatility in consumer spending and high industry concentration. The main challenge for retailers is to adapt their store mix to address changing consumer preferences.

Furniture, electronics, appliances, hobby and leisure: Segments with generally higher profit margins but greater volatility as they sell discretionary consumer goods. Fierce competition from e-commerce specialists. Uncertain transition from a brick-and-mortar to click-and-mortar business model.

Apparel and accessories: Much like other discretionary spending, apparel and accessories see reduced consumer spending when the economy decelerates.

Beauty and cosmetics: Very dynamic sales globally pushed by the retail expansion of luxury companies.

Department stores: Department stores face very strong competition from online stores. The segment is cutting capacities in North America and is starting to adapt in Europe, but still growing in other regions of the world.

E-commerce specialists: Buoyant top line growth but still elusive profitability for the vast majority of players.

Telecom

Space is the new limit or growth channel?

Sector rating

M Medium risk for enterprises

Strengths & weaknesses



- High entry barrier / infra difficult to replicate
- Mass-market business (=large and limitless B2C and B2B audience)
- Essential-utility services (critical for connecting people/activities each other)
- Long product shelf-life (recurring cash stream = better revenue predictability)
- Strong exposure to AI development
- (high real-data demand)



- Highly capital-intensive (heavy debt business)
- Strongly regulated (national authorities deliver spectrum licenses and set up price framework)
- Negative working capital (= slow monetization of large investments)
- Vulnerability of infra to climate events/ geopolitical tensions/social unrest
- Emerging competition from hyperscalers on satellite internet services

Sector overview

What to watch?

- Further consolidation of the sector through M&A, takeover bids, public auctions, JVs.
- Major advancements into 6G development/integration for B2C/B2B purpose and impacts on capex/free cash flow trendline.
- Increasing demand for real-time data driven by broader AI adoption (inv. slower construction pace for datacenters as key risk for the topline).
- Physical infra casualties caused by geopolitical conflicts/ climate events
- Regulatory framework on personal/industrial data usage and traffic (tighter rules for hyperscalers could mean more business for telecom operators)
- Revenue dilution risks from early development of LEO satellite-based broadband internet solutions



Will the AI boost be tangible in the telecom space?

Underlying demand is inelastic but barely growing: connectivity is a mature, saturated utility, and industry service revenue is expanding only ~1–1.5% while EBITDA rises ~5% on efficiency and cost discipline rather than volume. The genuine growth vector is the AI build-out, which is reshaping traffic and reviving enterprise/wholesale demand – operators can build and price for AI-era workloads through data-center interconnect, high-capacity backhaul and subsea capacity, and undersea-cable investment is booming specifically to serve the AI build. This is the sector's best organic story over the near future (12-24 months), but it is derivative: topline upside depends on the pace of hyperscaler data-center construction, so any slowdown in that capex wave feeds straight back into telecom wholesale revenue. The mix shift favors B2B, wholesale and infrastructure services over commoditized consumer connectivity – rewarding operators positioned as digital-infrastructure enablers rather than pure retail carriers.

Time for consolidation to fix fragmentation costs. After a decade of blocking in-market deals on consumer-price grounds, European authorities have pivoted toward scale: The Draghi framework reframed fragmentation as a competitiveness problem, and regulators are now described as more open to large-scale mergers that previously faced structural prohibition. The structural rationale is stark: The average European mobile operator serves ~4.5mn subscribers versus ~95mn in the US and hundreds of millions in India and China, leaving the region chronically sub-scale to fund 5G and fiber-based high speed connectivity solutions. Expect intensifying in-market M&A, takeover bids and JVs alongside continued carve-outs of towers, fiber altnets, subsea units and data centers. If consolidation restores pricing discipline, it is the single biggest lever for reversing the sector's revenue stagnation and margin drift; execution and integration risk are the offsets.

Goldilocks capex-relief era might soon end: 6G development under scrutiny. The heavy investment cycle has peaked: European capex intensity has fallen from a 5G-era high near 25% of revenue (2021) toward the mid-teens, after operators reached ~87% household 5G and ~70% FTTH coverage by end-2024. Falling capex is driving deleveraging – aggregate European net debt is easing toward ~2.5x EBITDA from a 2021–22 peak above 3x – though the sector remains structurally levered, with some North American majors still at ~3.5–3.9x, enforcing capital discipline and asset disposals. Solvency is therefore improving but not resolved; liquidity is underpinned by rich, dependable cash generation, as EBITDA margins hold above 35%. But the slow monetization problem persists – large upfront network and spectrum outlays convert to cash only over long horizons – and carriers see a looming overhang with 6G development, which is threatening to restart the capex cycle just as balance sheets recover, with no profitability guarantees nor concrete cashflow timeline on this investment case yet.

Geopolitical and regulatory clouds add to the pile of constraints. Two non-structural catalysts round out the outlook. First, physical vulnerability is now a board-level risk: around 200 subsea cables are cut or disrupted annually – mostly accidental – but geopolitically-linked incidents have clustered in the Baltic, around Taiwan and in the Red Sea through 2024–2025, straining infrastructure that carries up to ~99% of intercontinental data. Encouragingly, networks proved reroutable – recent cuts caused higher latency, not outage – but resilience, redundancy and security are becoming a structural opex and capex line, compounded by climate exposure. Second, the regulatory framework cuts both ways: spectrum licensing and price rules remain a structural constraint, and data-usage/traffic regulation and sovereignty requirements are tightening. Yet the same debate carries upside – the “fair contribution” argument, that hyperscalers generating the bulk of network traffic should help fund it, could translate into a new revenue stream if it advances. Regulation is simultaneously the sector's cost ceiling and, potentially, its next monetization avenue.

Space is becoming the new competitive frontier. Considering both constraints listed above likely to physically impact telecom infrastructure, the satellite dimension is where the limit or channel is genuinely undecided. On the threat side, LEO broadband is scaling fast and pricing aggressively: The dominant constellation has roughly 9–10mn subscribers (out of a total of around 12mn) across ~155 countries with near-monopoly consumer status, and head-to-head economics point to further price erosion as a hyperscaler-backed challengers enter. The direct revenue-dilution risk concentrates in rural, remote, maritime and aviation segments – precisely the coverage-gap and high-ARPU niches operators struggle to serve economically. But the same technology is increasingly a channel, not just a rival: Direct-to-cell partnerships let terrestrial operators extend coverage without building it, and carriers are signing satellite deals at scale – one alliance grants a satellite operator access to over 150mn potential mobile customers. Some operators partner with the incumbent constellation, others back alternative direct-to-device ventures to avoid dependence.

Subsectors

Landline and mobile telecommunications. The biggest players operate in both segments, but it is not the case in emerging economies.

Consumer and business telecommunications. Some niche players focus on the specific needs of corporates.

Cable companies.

Satellite telecommunications including New Space.

Mobile Virtual Network Operators (MVNO) are companies focusing on marketing mobile services while renting network capacity to traditional operators.

Textiles

Navigating rising costs and volatile consumer demand

Sector rating

 Sensitive risk for enterprises

Strengths & weaknesses



- Strong brand awareness as margin protector (image/representation > quality)
- Low capital intensity & agile business (easily reconfigured/relocated in case of disruption)
- Direct shipping business transiting from multichannel to omnichannel model
- Rising popularity of second-hand market increases product shelf-life and ROI
- Growing demand from B2B (technical fabrics) offsetting volatility in B2C segment



- Production costs sensitive to energy prices (synthetic fabrics = high chemical footprint)
- High reliance on China (directly or indirectly via high input footprint on external production)
- Rapid product obsolescence amid regular customer taste change (overproduction costs)
- Low entry barrier (low technology footprint) – high competition dragging down prices
- Rising compliance costs to absorb in Europe (recycling & local sourcing ration requirement)

Sector overview

What to watch?

- Raw material & production cost volatility amid new tensions in Middle East
- Consumer spending in non-essential durable goods as interest rates are rising again
- Long-term effects of tariffs on margins and production relocation outside Asia
- Growing demand from B2B segment, notably defense industry
- Integration of AI technology for automizing process and customizing customer journey
- Adaption of manufacturers/distributors to higher sustainability criteria in Europe
- Erosion of brand awareness due to product quality deterioration and persisting inflationary effects

Living in a more expensive world: margin and working capital deterioration. Raw materials account for roughly 70–85% of operating expenses across spinning and fabric production, so input moves land almost directly on gross margins. That burden has grown over the past two decades: as chemical-intensive synthetics have taken a larger share of global fabric output, so has exposure to oil prices. Meanwhile, more frequent extreme climate episodes – heatwaves and flooding – have degraded yields of textile commodities such as cotton, contributing to a lasting increase in production costs. Renewed Middle East tensions have pushed energy prices higher again in 2026, compounding the effect. Manufacturers cannot pass this through, at least not in full: fixed long-term contracts contain no mark-up provisions, and distributors resist meaningful price increases given demand elasticity outside luxury. The sector therefore absorbs input inflation rather than transmitting it – the defining margin mechanic of the window, and one that no commercial lever can correct.

Tariffs reshuffle the supply chain but not the geographic footprint: Asia keeps its edge. Tightening trade conditions add a layer of uncertainty for distributors and have revived the debate over relocation and sourcing policy. The timing is what makes this acute: apparel is committed six to nine months ahead, so orders must be priced against duty rates that may not exist when the decision is made. Diversification helps less than it appears – roughly a fifth to a third of Southeast Asian export value originates in China, with much of the fabric, investment and factory ownership Chinese, and flagged inputs triggering trans-shipment penalties.

The exposure was relocated, not removed, and the upstream layer is far harder to rebuild than cut-and-sew. Despite US tariffs, little production ultimately left the region: Vietnam replaced China as the largest apparel and textile supplier to the US in 2025 as established producers demonstrated genuine agility in reallocating asset-light capacity, preserving their competitive edge through an attractive expertise-to-cost trade-off. Nearshoring into duty-free, low-cost locations near Western markets remains idiosyncratic rather than a global trend – a complementary strategy buying distribution speed and predictability rather than decisive long-run cost savings.

Highly vulnerable to deferred consumption as inflationary pressure returns. Clothing is among the most postponable lines in the household budget, and weak appetite amid sluggish growth, persistent inflation and renewed rate pressure defines the window on both sides of the Atlantic. Brand equity is the principal margin protector, but it depreciates under exactly these conditions: inflation pushes consumers to test whether the premium is earned,

cost pressure tempts brands to thin specification behind unchanged prices and resale makes the durability trade legible by publicly pricing residual value. Beneath sits structural overproduction – output driven by revenue targets and capacity utilization rather than end demand, then cleared through discounting at minimal margin. With demand vulnerability compounding the margin squeeze, inventory agility becomes the decisive capability, since buffer stock held against tariff volatility adds carrying cost and markdown risk at elevated financing rates. This is where the only effective offset sits: AI-driven forecasting, real-time defect detection and on-demand small-batch production improve inventory turnover, capture demand shifts through live data and allow timely production adjustment. Because surplus stock and markdown are the sector's largest structural margin leak, precision here converts directly into realized gross margin and a shorter cash cycle – margin recovery in this window comes from inventory discipline, not price.

Solvency risk remains under close scrutiny. Low capital intensity is both the agility asset and the competitive curse: minimal entry barriers mean chronic fragmentation and no supply discipline when demand softens. The financial consequences are already visible – several hundred South Asian factories ceased operations over roughly two years, split between insufficient orders and financial strain, while European retail and consumer distress reached its highest level since the financial crisis in Q2 2026, with profitability the primary fault line. Solvency risk is therefore bilateral: brands face supply discontinuity, vendors face cancellation. Early adopters of inventory technology hold a clear advantage here, while laggards face profit decline and liquidity constraints converging into genuine insolvency risk. European regulation currently deepens the strain, having layered on successive compliance requirements covering traceability, recycled content and sourcing origin. Yet the same rules build the barrier: what reads as a cost today should convert into competitive advantage, consolidating positions in large consumer markets while providing a legal shield against competitors that have not met sovereignty and circularity standards.

Fast fashion erodes while technical fabrics and resale sustain momentum. The high-volume, low-price model is losing ground on both flanks – regulatory cost falls hardest on it, while value-seeking consumers increasingly satisfy the same demand through secondhand channels. Technical and industrial categories are forecast to outgrow fashion, with demand rising for antibacterial, fire-resistant and water-repellent fabrics across defense, healthcare, automotive and construction, alongside smart textiles integrating sensors

and conductive fibers. European defense procurement is the standout near-term driver: budget-backed, politically insulated and largely price-inelastic. For European producers it is also the most credible answer to Asian cost competition, because the barrier is technical qualification rather than labor cost. In parallel, tightening circularity requirements and persistent price-seeking behavior are expanding the secondhand market, which is expected to approach USD400bn globally by 2030 (vs. USD290bn estimated in 2026), growing at roughly twice the pace of the traditional retail channel.

Subsectors

Textiles: about 55% of industry turnover. Textile manufacturers mostly serve the needs of the clothing and home improvement industries, and to a lesser extent other industries (automotive, etc.)

Apparel: about 30% of industry turnover. Apparel manufacturers are mostly contractors to which large fashion retailers outsource the actual manufacturing of their clothes and accessories.

Leather and shoes: about 15% of industry turnover. Leather goods and shoes generally have distribution channels that are distinct from apparel.

Asia Pacific concentrates 80% of the industry's turnover, with China's share itself standing almost 65%, the remainder being split between Cambodia, India, Indonesia, Bangladesh and Vietnam. Excluding Asia, Turkey (3%) and Brazil (1%) also have sizeable textile industries exporting to the European and US markets, respectively. Some countries producing high-end or technologically advanced fabric have kept a significant manufacturing base (Italy, France, Germany, South Korea and the US in particular).



Transport Equipment

Facing headwinds: Economic and geopolitical forces reshaping the industry's dynamics

Sector rating

 Sensitive risk for enterprises

Strengths & weaknesses

- 
- Strong pricing power, with limited number of companies capable of designing, certifying and manufacturing
 - High barriers to entry.
 - Robust long-term demand outlook, supported by fleet renewal, rising defence expenditure, infrastructure investment and the transition towards lower-emission transport.
 - Long order backlogs.
 - Stable customer relationships.
 - Long production cycles that facilitate production planning, capacity utilization and financial visibility.
 - Significant aftermarket and maintenance revenues, generating recurring, high-margin cash flows over the product lifecycle.
 - Continuous technological innovations.
 - Improving supply-chain resilience.

- 
- High capital intensity, requiring substantial and continuous R&D investment.
 - Elevated working capital requirements, driven by lengthy production cycles, large inventories and milestone-based customer payments.
 - Generally high financial leverage, particularly during periods of major investment programs or production ramp-ups.
 - Exposure to raw material and commodity price volatility.
 - Stringent regulatory, certification and safety requirements, increasing development costs, extending time-to-market and exposing manufacturers to costly delays.
 - Long development cycles and program execution risk, with significant financial consequences from production delays, cost overruns or technical issues.
 - Cyclical exposure to global economic conditions, especially in commercial aerospace, freight transportation and heavy-duty trucks.
 - Dependence on complex global supply chains.
 - Labor shortages and skills constraints.
 - Geopolitical and trade risks, which can affect supply chains, market access and procurement decisions.

What to watch?

- **Defense spending becomes a structural growth driver:** Rising geopolitical tensions continue to support strong demand for military aircraft, naval vessels and combat vehicles. Multi-year procurement programs are improving revenue visibility for defense-oriented OEMs, although rapidly expanding production capacity remains a key execution challenge.
- **Production ramp-ups and supply-chain execution:** Supply chains have largely normalized following the pandemic, but commercial aerospace continues to face bottlenecks in engines and critical components. The industry's focus has shifted from generating demand to increasing production rates while maintaining quality and controlling costs.
- **Decarbonization drives long-term investment:** Stricter environmental regulations continue to accelerate investment in cleaner technologies, including sustainable fuels, electrification and hydrogen propulsion. While disruptive technologies remain a long-term opportunity, manufacturers are currently prioritizing incremental efficiency improvements and lower-emission solutions.
- **Commodity and energy price volatility:** Although commodity prices have stabilized compared with recent peaks, volatility in metals, critical minerals and energy continues to pressure manufacturing costs. Profitability remains sensitive, particularly under long-term fixed-price contracts.
- **Labor shortages and industrial capacity:** Demand for skilled engineers and technicians continues to outpace supply, limiting production ramp-ups across several segments. Persistent wage inflation and talent shortages remain key operational challenges.
- **Industrial policy and trade tensions:** Governments are increasingly promoting domestic manufacturing through subsidies, local-content requirements and trade measures. While these policies strengthen supply-chain resilience, they also increase costs and add complexity for globally integrated manufacturers.
- **Strong order backlogs:** Record order books across aerospace, rail and naval markets provide several years of production visibility and support long-term investment. However, converting these backlogs into revenue depends on successful program execution and supplier performance.
- **Digitalization and automation:** Manufacturers are increasingly leveraging AI, digital twins and advanced automation to improve engineering, optimize production and reduce lifecycle costs. Digital capabilities are becoming an increasingly important competitive differentiator.

- **The expanding space economy:** Growth in commercial launch services, satellite constellations and lunar exploration continues to create new opportunities for aerospace manufacturers. Although still relatively small, the space market remains an attractive long-term source of innovation and technological development

Overall challenges influencing market dynamics: The transport equipment sector continues to experience contrasting dynamics in 2026, with strong long-term demand supported by defense spending, fleet renewal and infrastructure investment, but constrained by execution challenges and macroeconomic uncertainty. While many manufacturers benefit from historically high order backlogs, the conversion of these orders into revenues remains dependent on supply-chain reliability, production capacity and cost control. At the same time, higher trade barriers, industrial policy shifts and geopolitical tensions are reshaping global manufacturing strategies. Increased localization requirements and tariff uncertainty may delay investment decisions in some end markets, while rising labor, energy and raw material costs continue to pressure margins, particularly for manufacturers operating under fixed-price contracts.

Aviation outlook: The aviation sector remains supported by robust underlying demand, driving record-high backlogs, but continues to face significant production constraints. Aircraft manufacturers continue to face challenges in ramping up output due to shortages of critical components, limited supplier capacity and delays across the aerospace supply chain, particularly in engines, aerostructures and specialized materials. As a result, aircraft deliveries remain below manufacturers' previous targets. The industry is also undergoing a major technological transition driven by decarbonization objectives. Investments in sustainable aviation fuels (SAF), lightweight materials, more efficient engines and longer-term solutions such as electric and hydrogen propulsion are accelerating, although large-scale adoption of disruptive technologies remains a long-term prospect. Going forward, the key challenge for aircraft manufacturers will be balancing record demand with operational execution: increasing production rates while maintaining strict safety standards, managing supplier constraints and avoiding further program delays.

Shipbuilders outlook: The shipbuilding industry is undergoing a significant strategic transformation driven by geopolitics, decarbonization and changing trade patterns. While China remains the dominant global shipbuilding hub due to its scale and cost competitiveness, growing geopolitical tensions and efforts to reduce supply-chain dependence are encouraging operators to diversify

sourcing towards South Korea, Japan and other emerging shipbuilding markets. Environmental regulations are also reshaping demand, with shipowners investing in new vessels capable of meeting stricter emissions requirements. This is supporting demand for LNG-powered vessels, alternative fuel technologies and more energy-efficient designs. However, the transition requires substantial investment and increases technological complexity for shipbuilders. Despite healthy order activity in several segments, profitability remains sensitive to raw material costs, labor availability and competitive pressure. The industry is therefore entering a period of structural realignment, where geopolitical considerations are becoming increasingly important alongside traditional cost competitiveness.

Rolling stock manufacturers outlook: The global rail equipment market benefits from strong structural tailwinds, supported by urbanization, infrastructure investment, decarbonization efforts and the shift towards lower-emission transport. China is by far the largest rail market, driven by extensive high-speed rail expansion and urban metro development, while Europe and Japan remain global technology leaders in high-speed rail, signaling and advanced rolling stock. India, the Middle East and Southeast Asia represent the fastest-growing demand markets, supported by major infrastructure investment programs and urban mobility needs. As of today, the sector faces execution challenges, including production capacity constraints, supply chain pressures, labor shortages and lengthy certification processes, which can delay project delivery and weigh on margins. Profitability remains pressured by inflationary cost increases and competitive bidding environments, while rising competition from Chinese manufacturers in international tenders represents a growing challenge for established European and Japanese players. At the same time, digitalization, automation and higher-value aftermarket services provide opportunities to improve long-term profitability.

Commercial vehicles outlook: Commercial vehicle manufacturers face a more challenging environment due to weaker economic growth, softer freight demand and uncertainty surrounding the transition towards lower-emission technologies. After strong post-pandemic demand, several major markets, particularly Europe, have experienced declining truck and van registrations as high financing costs, weaker industrial activity and fleet replacement delays weigh on demand. At the same time, manufacturers are facing significant investment requirements to comply with stricter emissions regulations and accelerate the transition towards electric and alternative-fuel vehicles. The heavy-duty segment remains particularly challenging, given infrastructure limitations, high battery costs and uncertainty

regarding the optimal long-term technology pathway. The sub-sector in Europe is also exposed to increasing competitive pressure, particularly from Chinese manufacturers that are expanding internationally with cost-competitive electric commercial vehicles. Going forward, the main challenge for OEMs will be balancing profitability with the need to invest heavily in electrification, digitalization and autonomous driving technologies.

Subsectors

Aircraft manufacturers: Companies involved in the design, engineering and assembly of any kind of vehicle that is manufactured to fly in the air. This includes planes (commercial and cargo planes, private jets, military jets and propeller planes) and helicopters (civil and military). The US and France are the largest producers of aircraft in the world.

Shipyards: Companies that build and repair any type of water transport vehicle, such as yachts, cruise ships, container and dry bulk vessels, military vessels, oil and chemicals tankers, ferries, ice-breakers, etc. By revenue, this market is vastly dominated by South Korean, Chinese and Japanese companies, with European shipyards following after the Asian ones.

Rolling-stock manufacturers: Companies involved in the entire process of designing, manufacturing, assembling and testing rolling stock. Wagons/trains can be used for both passenger and cargo transport and can be high-speed or not. This market is led by China and Europe.

Trucks OEMs: Manufacturers of trucks or camions, which are normally used for cargo transportation or for carrying other vehicles or dry bulk commodities. Trucks vary in size, power and configuration. Today, most of them are still powered by diesel fuel, although electric trucks have slowly been introduced to the market.



Transportation

Building resilience in an era of disruption and transformation

Sector rating

S Sensitive risk for enterprises

Strengths & weaknesses



- Essential economic infrastructure: Transport is a critical enabler of trade, industrial activity, tourism, and labor mobility, making it indispensable to economic development.
- High barriers to entry: Significant capital requirements, regulation and infrastructure needs protect incumbents and limit new competition.
- Resilient long-term demand: Demand is mostly supported by structural trends such as globalization, urbanization, population growth and e-commerce, despite cyclical fluctuations.
- Market concentration: Several transportation segments exhibit monopolistic or oligopolistic characteristics, supporting relatively strong pricing power and stable competitive positions.
- Cost efficiency and economies of scale: Maritime shipping and rail remain among the most efficient transport modes for moving large volumes over long distances.
- Public-sector support: Strategic transportation infrastructure often benefits from government funding, subsidies and favorable policy frameworks.
- Digital transformation: AI, automation and data analytics are improving operational efficiency, asset utilization and customer service across the sector.



- Capital-intensive business model: High investment and maintenance requirements result in substantial fixed costs and long asset replacement cycles.
- Financial and commodity exposure: High leverage and sensitivity to fuel and energy prices can significantly affect profitability and cash flow.
- Cyclical demand: Freight volumes and passenger traffic are closely linked to economic activity, trade flows and consumer confidence.
- Decarbonization pressures: Stricter environmental regulations and the transition to low-emission technologies require substantial ongoing investment.
- Labor constraints: Persistent shortages of skilled workers across several transport modes continue to increase operating costs and create operational challenges.
- Geopolitical and operational risks: Trade disputes, conflicts, supply-chain disruptions and cybersecurity threats can materially affect operations and profitability.
- Infrastructure dependence: Performance depends on the quality, capacity and public investment in transport infrastructure, which may constrain growth and efficiency.

Sector overview

What to watch?

- Geopolitical fragmentation and the reshaping of global trade flows: Rising geopolitical tensions, protectionism and supply-chain vulnerabilities are fundamentally reshaping transportation networks. Companies are increasingly prioritizing resilience over pure cost optimization, driving supply-chain diversification, nearshoring and the development of alternative trade

corridors. Maritime shipping remains particularly exposed due to dependence on strategic chokepoints such as the Suez Canal, the Strait of Hormuz and the Panama Canal, while aviation continues to face disruptions from restricted airspace and geopolitical tensions. These trends are likely to result in longer routes, higher logistics costs and increased investment in more flexible and resilient transportation networks.

- **Decarbonization and the transition toward low-carbon transportation:** Transportation remains one of the largest contributors to global greenhouse gas emissions, making it a central target of regulatory and investor pressure. Stricter emissions standards, carbon-pricing mechanisms and sustainability requirements are accelerating fleet renewal and the adoption of alternative technologies. However, decarbonization remains particularly challenging for aviation and maritime shipping due to technological limitations, long asset lifecycles and the limited availability of competitive low-carbon fuels such as sustainable aviation fuel (SAF) and alternative maritime fuels. Companies able to secure access to cleaner technologies and fuels may gain a significant competitive advantage.
- **Energy transition and long-term fuel uncertainty:** Transportation is entering a period of significant energy transformation, with uncertainty surrounding the future fuel mix across different modes. While EVs are expanding rapidly in passenger and short-haul applications, heavy-duty road transport, aviation and maritime shipping face greater technological challenges. Continued volatility in oil prices, combined with increasing demand for electricity and alternative fuels, will influence operating costs, investment decisions and competitive positioning across the sector.
- **AI, automation, and the transformation of transportation models:** Rapid advances in AI, automation and digital technologies are reshaping how transportation assets are operated and managed. Predictive maintenance, autonomous vehicles, intelligent logistics platforms and real-time optimization are improving efficiency and reducing costs. However, the uneven adoption of these technologies may create a widening gap between technologically advanced operators and traditional players. Cybersecurity risks will also increase as transportation systems become increasingly connected and data-dependent.
- **Labor availability and the evolution of transportation workforce requirements:** Persistent shortages of skilled workers remain a structural challenge across trucking, aviation, maritime and logistics operations. An aging workforce, demanding working conditions and rising wage pressures are increasing operating costs and limiting capacity growth. At the same time, automation and digitalization are changing workforce requirements, increasing demand for employees with technical, analytical and digital capabilities while reducing reliance on certain traditional roles.
- **Infrastructure constraints and investment gap:** Aging infrastructure, congestion and insufficient public

investment may become increasingly significant constraints on transportation growth. Ports, airports, roads and rail networks require substantial modernization to accommodate growing demand, improve resilience and support decarbonization. The ability of governments and private operators to finance these investments will influence future sector competitiveness.

Airlines: The airline industry has largely completed its post-pandemic recovery, with passenger demand returning to a more normalized growth trajectory. But after reaching record levels in 2025, global traffic has softened in 2026 amid weaker economic growth and heightened geopolitical uncertainty. Industry-wide revenue passenger kilometers (RPK) declined by -1.7% y/y in June (after a -2.2% decline recorded in May), while the passenger load factor (PLF) remained resilient at 84.2%, reflecting disciplined capacity management despite softer demand. Air cargo has followed a similar trend, with demand remaining above pre-pandemic levels but expanding at a more moderate pace.

The industry's main challenge in 2026 has been the escalation of geopolitical tensions in the Middle East.

Airspace closures and security concerns have disrupted one of the world's most important aviation hubs, forcing airlines to reroute flights between Europe and Asia, increasing flight times, fuel burn and operating costs while weighing on regional tourism. Middle Eastern carriers have been the most affected, with passenger traffic remaining around 14% below last year's levels despite a gradual recovery since the peak of the disruptions. Although global capacity is expected to resume modest growth during the last quarter of 2026, airlines are likely to remain highly exposed to further geopolitical developments.

Profitability is also being squeezed by record jet-fuel prices, which have become the industry's most important margin headwind. Unlike crude oil, kerosene prices have surged due to elevated refining margins and disruptions to Middle Eastern fuel supply chains, with fuel accounting for roughly 30% of airlines' operating costs. The impact varies across regions: most US airlines remain largely unhedged against fuel prices but have partly offset higher costs through strong travel demand associated with the FIFA World Cup, while European carriers entered the year with stronger fuel hedges but remain more exposed to disruptions in Middle Eastern jet-fuel supplies and longer flight routings around restricted airspace. Combined with higher labor, maintenance and airport costs, airlines are expected to operate in a structurally higher-cost environment than before the pandemic.

Looking ahead, we expect industry revenues and passenger demand to continue growing at a modest pace through the remainder of 2026 and into 2027, broadly in line with global economic activity. Profitability should gradually improve as aircraft deliveries recover and geopolitical disruptions ease, although margins are likely to remain below historical peaks given persistently high operating costs, the ongoing transition toward sustainable aviation fuels and continued uncertainty surrounding global fuel markets.

Maritime: Maritime shipping has transitioned from a relatively predictable backbone of global trade into a structurally more volatile and risk-exposed industry. Geopolitical tensions, trade fragmentation, climate-related disruptions and increasing regulation are reshaping global shipping routes and raising operating complexity. The Red Sea crisis remains the clearest example: continued security risks around the Bab el-Mandeb Strait have forced many carriers to reroute vessels around the Cape of Good Hope, extending voyages by approximately 35–45%, increasing fuel consumption, insurance costs and fleet requirements. More recently, tensions around the Strait of Hormuz have reinforced concerns about the vulnerability of strategic maritime chokepoints, particularly given its importance for global oil and LNG flows.

Freight rates have normalized significantly from the exceptional levels observed during the 2021–2022 supply chain crisis, when container prices reached peaks of around USD15,000 per FEU on certain routes. However, volatility remains structurally higher than before the pandemic due to recurring disruptions, shifting trade patterns (the US tariffs war in 2025 lifted rates up) and uncertainty around global supply chains. In addition, the large wave of vessel orders placed during the post-pandemic boom is now increasing available capacity, creating a risk of overcapacity.

At the same time, the industry is entering a long-term investment cycle driven by fleet renewal and decarbonization needs. Stricter environmental regulations, carbon pricing mechanisms and the need to replace an aging global fleet are accelerating demand for new vessels and alternative fuels. While this creates opportunities for operators with modern, efficient fleets, it also represents a significant financial challenge given the high cost of new ships and uncertainty around future fuel technologies.

Looking ahead to 2027, maritime transportation is expected to remain essential for global trade but operate under a “new normal” characterized by higher volatility, greater capital intensity and increased geopolitical exposure.

Profitability is likely to remain under pressure as fleet capacity expands and bunker fuel costs stay elevated due to continued uncertainty around Middle Eastern energy flows. Higher fuel expenses, longer sailing routes caused by geopolitical disruptions and increasing decarbonization requirements will continue to weigh on margins, particularly for operators with older and less efficient fleets. However, companies with strong balance sheets, modern vessels, diversified networks and early positioning in low-carbon technologies should be better positioned to outperform in an increasingly complex operating environment.

Rail: Rail transport continues to strengthen its strategic position as governments increasingly prioritize low-carbon and energy-efficient mobility. Across Europe, public investment remains focused on expanding high-speed rail, modernizing existing networks and improving cross-border connectivity through projects such as the Trans-European Transport Network (TEN-T). At the same time, the gradual liberalization of passenger rail markets is increasing competition, encouraging service innovation and improving operational efficiency, particularly on high-demand intercity routes.

Compared with other transport modes, rail has been relatively insulated from the sharp increase in fuel prices during 2026. The widespread electrification of European rail networks has reduced direct exposure to oil markets, making rail an increasingly attractive alternative for both passengers and freight as airlines and road operators face higher fuel costs. At the same time, rail continues to benefit from governments’ decarbonization strategies, with subsidies and public funding supporting network expansion, fleet modernization and electrification. Looking ahead, we expect rail to continue gaining market share on short- and medium-distance corridors, supported by environmental policies, sustained public investment and growing demand for sustainable mobility. The main challenges remain infrastructure capacity constraints, aging networks in several countries and the need to accelerate digitalization and signaling upgrades. Overall, rail is likely to remain one of the transportation sector’s most resilient segments through 2027.

Road: Road transportation continues to benefit from sustained government investment in infrastructure, particularly in developed markets where modern road networks are viewed as strategic assets for strengthening economic competitiveness and supply-chain resilience. Major initiatives such as the EU’s TEN-T programme and the US Infrastructure Investment and Jobs Act are expected to support freight activity and passenger mobility through

2027, while inadequate infrastructure remains a structural constraint across many emerging economies.

The sector's main challenge in 2026 has been the sharp increase in diesel prices following the escalation of the conflict in the Middle East. As most road transport operators have limited fuel-hedging capabilities, higher fuel costs have translated rapidly into margin pressure, with only partial recovery through fuel surcharges in competitive freight markets. At the same time, governments continue to accelerate fleet electrification through subsidies and charging infrastructure investments, although high vehicle costs and insufficient charging networks remain key barriers for heavy-duty transport. Looking ahead, we expect the sector to deliver modest growth through 2027, supported by infrastructure spending and resilient freight demand. However, profitability will remain closely linked to fuel-price volatility and the pace of the transition toward low-emission transport.

Subsectors

The transportation sector encompasses all kind of companies providing the means for transporting people and goods from one geographical place to another.

Air transport: Companies that use aircraft (owned or leased fleet) as a means of transportation, regardless of the distance (short or long-haul).

- a. Traditional airlines.
- b. Low-cost airlines: They offer the same transport facility of traditional airlines but with a reduced offer of services that enables lower costs, and therefore lower air-ticket prices.

Maritime transport: All kind of transportation services through boats, including:

- a. Cruise ships: large passenger ships used mainly for vacationing purposes.
- b. Vessels: container ships, oil tankers, chemical tankers, gas and LNG tankers, dry bulk carriers, car carriers (ro-ro ships).
- c. Ferries: mainly for short trips across rivers, harbors and channels or between islands.

Rail/train transport: Another way for moving people and goods over short and long distances (both underground and over the surface). Train systems run on metal rails, which allows the rolling stock to benefit from lesser frictional resistance and therefore to attach more load in terms of wagons or carriages. Besides being the means of transport that emits the least CO₂, it is also the one that is less affected by weather turbulence.

Road transport: This segment includes companies that offer the movement of people and goods through trucks, buses, coaches and taxi-cabs, both in urban perimeters and on highways.



Sector risk methodology

The Sector Risk Rating by Allianz Trade Economic Research assesses the risk of non-payment by companies in 17 sectors across 70 countries around the world. It is measured on a four-level scale from Low to High.

Sector risk assessments are based upon the forward-looking evaluation of four key determinants – demand, profitability, liquidity and business environment – using Allianz Trade internal data and expert judgments, as well as hard data from secondary sources.

Our grading system uses a unique methodology which combines data and expert judgments to assess the risk of nonpayment at a sector level across the top 70 countries of the world.



4 levels of risk based on 4 key components

The Sector Risk Rating is based on the evaluation of four components that are analyzed globally for each sector

Demand: Outlook for companies' turnovers based on the organic growth, fundamentals and price competition of the sector

Profitability: Outlook for companies' margins and profits depending on the evolution of prices in raw materials/ commodities, on labor costs and fluctuations in supply and capacity

Liquidity: Outlook for companies' cash positions and financing risks, based on access to financing and payment performance, and

Business environment: Any technological innovations, new government subsidies and changes in legal framework that can alter business models and companies' strategies.

A quantitative and qualitative grading system

Our grading system is a unique combination of indicators and expert judgements dedicated to assessing the short-term outlook of the four sub-components of our sector risk ratings.

The indicators are based on Allianz Trade internal data and hard data from secondary sources.

The expert judgments capitalize on the microeconomic expertise of Allianz Trade credit analysts, who closely monitor risk in companies all over the world, and the sector advisors of the Economic Research team, who analyze industry trends globally.

These judgments are collected using a standardized and consistent quarterly questionnaire.

Sector risk ratings are designed to complement Allianz Trade' Country Risk ratings and individual buyer risk assessments.





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Central Bank Credibility as a risk: Why the ECB leads, the BoJ lags and the Fed and BoE are scrutinized

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In Summary

Central bank credibility is becoming a risk rather than a background condition. At Jackson Hole, the new Fed Chairman called for a quieter central bank, breaking with two decades of managing expectations while the US Treasury is co-steering financial conditions through debt maturity and buybacks. Meanwhile in Europe, cancelling debt held by the Eurosystem has entered the French presidential campaign and investors still remember the Bank of England's emergency bond purchases during the Mini-Budget crisis. Against geopolitical tension, partisan polarization, inflation overshooting and elevated sovereign debt, these are all risks for central bank credibility – long a shock absorber for markets, it may start amplifying shocks instead.

We introduce a framework measuring credibility on six dimensions: the ECB tops the ranking, the BoJ carries the highest overall risk, and the Fed and BoE each show different vulnerabilities. We use six dimensions, three are structural: *institutional credibility* (legal independence, leadership stability), *expectation credibility* (how memory of past inflation shapes expectations) and the *credibility gap* (control over long-term yield trend); and three are cyclical: *credibility risk sentiment* (media coverage questioning independence), *market credibility* (how well prices and surveys track the inflation target) and *credibility pressure* (difficulty defending the price of money). The ECB ranks highest, an asset for the euro's international role. The Fed is strong structurally but weaker cyclically; the BoE's weakness is structural. The BoJ carries the highest risk, hit by JPY depreciation and long-end steepening at once.

Two trends capture the erosion: news-based concern is at its worst since 2017, and central banks are judged on a shorter track record than before 2022. The Fed drives 70% of global news coverage on central-bank credibility, yet Americans are less worried than the rest of the world. The opposite holds for the ECB, BoJ and BoE, where concern is mainly domestic. In parallel, we estimate how many years of past inflation shape expectations – how long the public's memory of past mistakes lasts. That memory fell from eight years to four in the 1980s disinflation fight, rebuilt to an 8-12-year peak by 2021, then was cut toward early-2000s levels by 2022. Credibility holds on market and survey measures, but long-earned trust has been in part lost.

Central banks have not regained their grip on the long-term yield trend after ending QE, opening a credibility gap. For the Fed and BoJ, yields sit above what policy implies – closing that gap may require buying long-term bonds again. For the ECB and BoE this is less urgent. Each gap opened at a distinct balance-sheet turn: end of QE for the Fed, end of yield-curve control for the BoJ, start of QE for the ECB and BoE. Markets seem to underprice renewed bond-buying odds in the US and Japan.

Restoring credibility means acting on four fronts, the cheapest being the most urgent: publish a rule-based framework with voting records so giving up forward guidance reads as discipline, not silence; anchor communication in reference points like wage growth or nominal GDP; define the Treasury relationship before issuance becomes a rival lever on rates and, most urgently, decide now how to respond if rates hit zero again, rather than improvise under the next shock. Credibility once let central banks avoid intervening at all; reversed, it will amplify shocks instead of absorbing them – and the Fed and BoJ have the least room to disappoint.

Central bank credibility moved from background condition to a risk factor

To establish which of the G4 (Fed, ECB, BoJ, and BoE) is most exposed to credibility risk, we built the a framework combining six measures. On the structural side we analyze the institutional setting (Institutional Credibility), monetary policy expectations (Expectation Credibility) and the capacity to anchor the long-term yield trend (Credibility Gap). As cyclical measure we propose the news sentiment (Credibility Risk Sentiment), market and survey indicators and the price of money (Credibility Pressure). We focus on the price stability target and policy rate setting, which the four share, recognizing that some carry a dual mandate or additional financial stability responsibilities.

The institutional framework: between *de jure* and *de facto* independence

Legal central bank independence is at a peak globally, but is mainly an European phenomenon. From a legal point of view, their independence has never been stronger. Over the last 50 years, this independence has been increasingly enshrined in the constitutional framework of developed and emerging countries. The main push did not come with the monetarist revolution of the 1980s. While the subordination of monetary to fiscal policy was identified at the time as a reason for the ineffectiveness of central banks during the 1970s inflation shock, this did not lead to a legal (*de jure*) separation of political and monetary decision-making. Instead, the independence of monetary policy came about through a targeted board appointment policy, which led to a change in economic culture and, ultimately, to a form of informal *de facto* institutionalization. The biggest global push in *de jure* independence came in the 1990s with the creation of the Eurozone, when future member countries adjusted to the standard of the German Bundesbank. One could even say that elevated *de jure* central bank independence is a largely European phenomenon. The US, UK, Japan, Australia and most emerging countries have only slightly tightened their legal frameworks over time. Their independence mainly relies on *de facto* elements. Among the G4 central banks, institutional credibility is weakest for the BoJ and BoE (Figures 1 & 2).

Figure 1: *De jure* central bank independence conceived in the 1980s and born in the 1990s

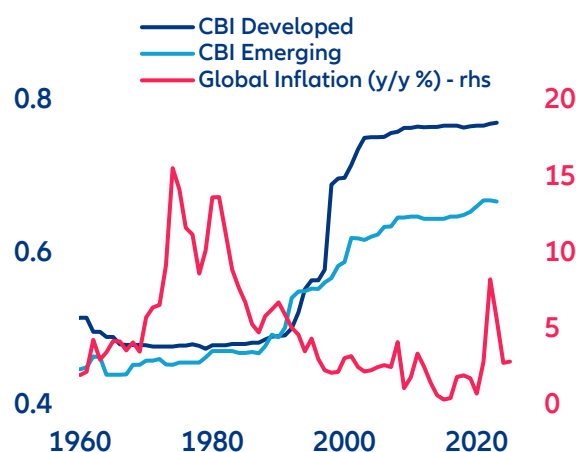
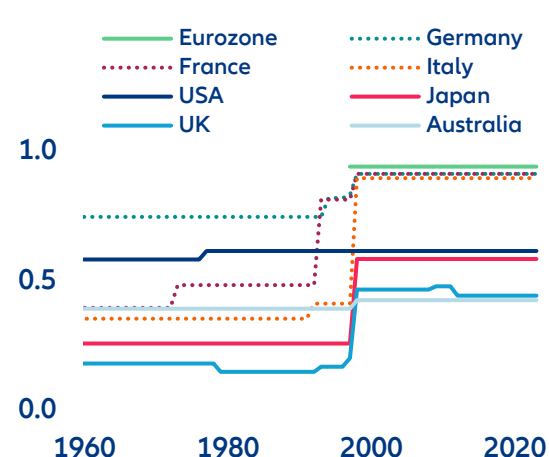


Figure 2: Strong *de jure* independence is a largely European phenomenon



Note: Central Bank Independence Index (CBI Index): 0 = no legal independence; 1 = maximum legal independence

Sources: CBI Database by Romelli (2025), LSEG Datastream, Allianz Research

When measuring the credibility risk for central banks arising from their *de jure* framework we use the Central Bank Independence Index (CBI Index)¹. But since many central banks rely on a *de facto* model of independence, we enhance the index by increasing the weight of nomination policy and add elements that reflect the risks of political interference, such as the number of irregular changes at the board and the quality of government. The ECB and the Fed lead this ranking, with the ECB being tighter on the legal framework while the Fed shows stronger institutional continuity. Compared to the purely *de jure* approach, the

¹ Romelli, D. (2025)

introduction of *de facto* elements decreases the independence score for the ECB while it increases the score for the Fed, the BoE and the BoJ (Figures 3 & 4).

Figure 3: *De facto* central bank Independence

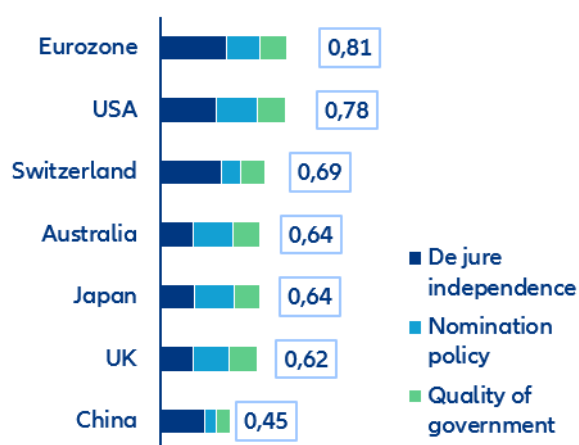
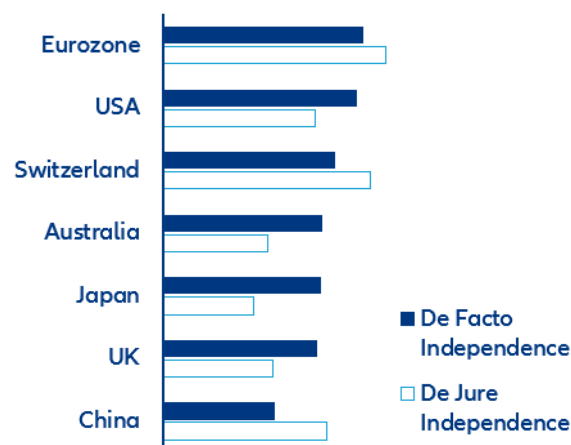


Figure 4: *De facto* vs *De jure* Independence



Note: *de facto* score: 50% CBI Index database with a double weight on "Governor and Central Bank Board policy" subindex, 30% ETH Zürich (KOF) Governor Turnover database and 20% Quality of Government (QoG) score from University of Gothenburg
Sources: CBI Database, ETH Zürich (KOF), University of Gothenburg, Allianz Research

Central bank sentiment signaling elevated level of credibility risk

News-based central-bank credibility has entered a new era of concern. *De jure* and *de facto* independence are necessary elements, but not sufficient to create central-bank credibility. A central bank strongly independent on paper can lack credibility, while a less legally independent one may retain market confidence if its actions are perceived by households, corporates and investors as effective. We measure this sentiment by tracking the occurrence of the words "independence" and "credibility" in combination with "central banks" and other monetary policy concepts in the global news flow and determine the sentiment attached to it (using the GDELT database). We see that the awareness (percentage of global news flow) around central-bank credibility has surged since 2025 and is associated with a cyclical low in sentiment expressing elevated credibility concerns. The volatility of the sentiment and its partial disconnect with awareness is also remarkable. Previously, sentiment moved in cycles mirroring public awareness: low awareness coincided with improving sentiment, rising awareness with sentiment deterioration (Figure 5).

Figure 5: Central-bank credibility peaks in global news with a strongly negative sentiment Z-Scores of frequency and sentiment

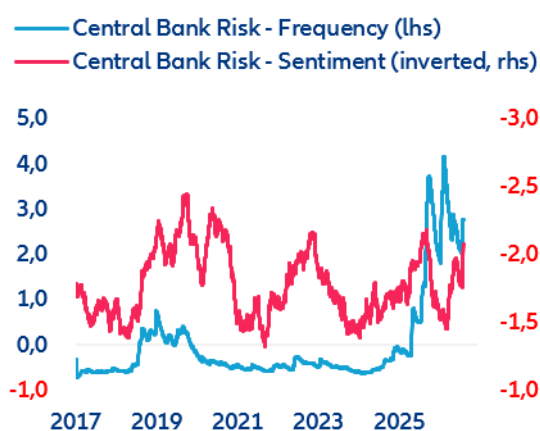
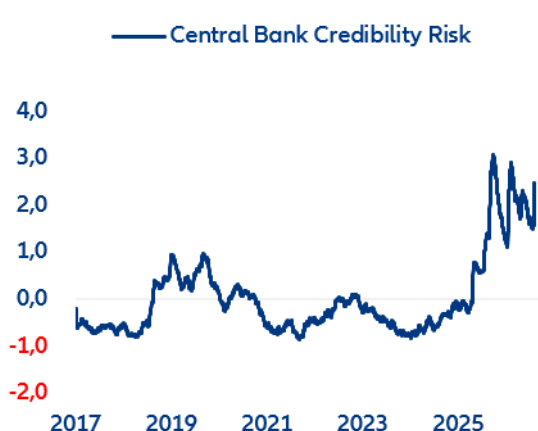


Figure 6: Credibility Risk Sentiment entered a new era of global concern
Z-Score Global Central Bank Credibility Risk



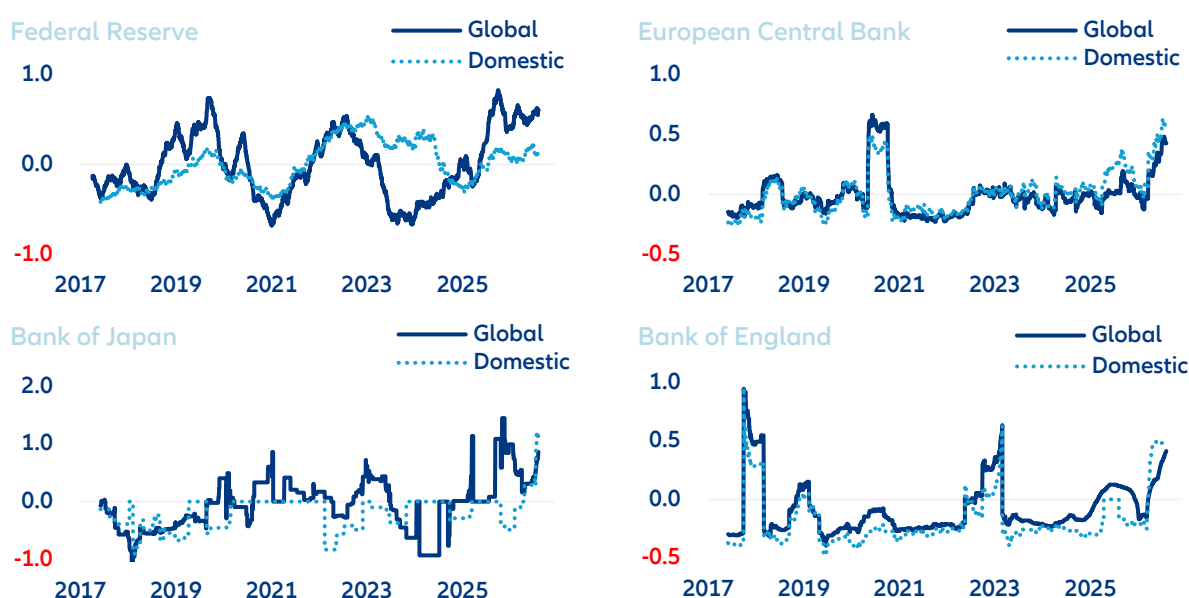
Note: data derived from GDELT news database measuring frequency and sentiment of word "credibility" and "independence" in combination with the terms "central bank", "Fed", "Federal Reserve", "ECB", "European Central Bank", "BoJ", "Bank of Japan", "BoE", "Bank of England", "monetary", "monetary policy", "inflation target", "fiscal dominance", "price stability" and "mandate"
Sources: GDELT, Allianz Research

When combining awareness and sentiment into one global Credibility Risk Indicator, we see that we have entered a new era – central-bank credibility has never been questioned as much as it is being questioned now (Figure 6).

The Fed suffers from an international credibility deficit, while for other central banks credibility concerns are stronger domestically. The global index is naturally dominated by US developments. Fed-related news flow accounts for roughly 70% of the global index. But the same trend holds when we examine individual central banks' credibility risks. The G4 (Fed, ECB, BoJ, and BoE) all stand at or near historical highs, with the BoJ being the most challenged in terms of credibility, followed by the Fed. Comparing domestic with global sentiment, the Fed's domestic news flow reflects less credibility risk than on the global level. The reverse holds for the ECB, the BoJ and the BoE, where domestic sentiment expresses more credibility concerns than global perception. This may relate to the global role of the USD: international investors react more sensitively to news flow here, while for the other central banks the domestic focus remains predominant (Figure 7).

Figure 7: Credibility Risk for different central banks – domestic vs global perception

Z-Scores of domestic and global Central Bank Credibility Risk



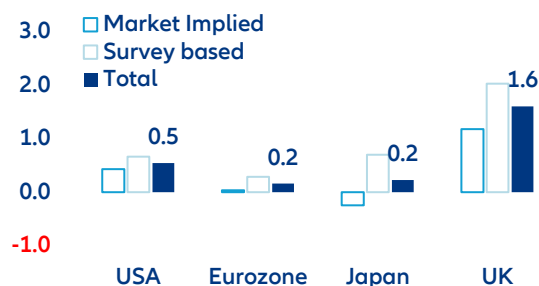
Sources: GDELT, Allianz Research

Inflation expectations from markets and surveys: a convenient but incomplete measure of central-bank credibility

The ECB and the Fed contain market- and survey-based inflation measures more successfully, which suggests lower credibility risk in the eyes of private sector agents. The conventional approach to quantifying central-bank credibility in terms of private-sector inflation expectations draws on a set of market-implied indicators such as breakeven inflation rates, inflation swap rates and inflation forward swap rates, alongside survey-based long-term expectations. It assesses their stability and their distance from the inflation target. The 5y5y inflation swap and the University of Michigan consumer inflation expectations have, in particular, acquired quasi-official standing as the market's and households' reading of long-run inflation anchoring. Combining their volatility, their deviation from the central-bank target since 2006 and the duration of target under- and overshooting, our analysis shows that the ECB and the Fed face less credibility risk since they contain these expectation indicators. The BoJ and the BoE, on the other hand, struggle on the back of extended episodes of over- and undershooting (Figures 8 to 11).

Figure 8: Deviation of long-term expectations from inflation target since 2006

Average deviation in pps



Notes: Market-implied: 5y5y inflation swap, 10y inflation swap, 10 breakeven inflation, Surveys: NY Fed, Cleveland Fed, ECB, BoJ, BoE, University of Michigan

Figure 9: Volatility of long-term inflation expectations

Standard deviation in pps, last 20 years

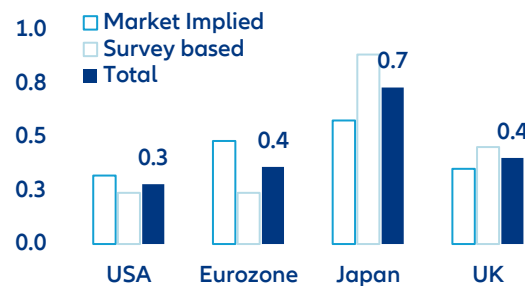
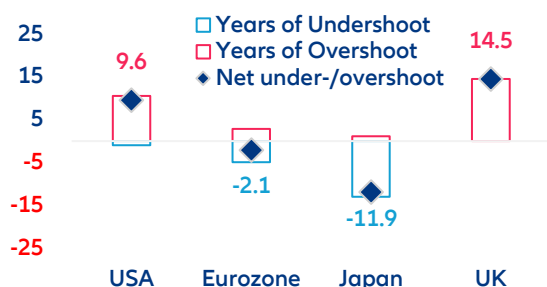


Figure 10: Years of long-term expectations under-/overshooting inflation target

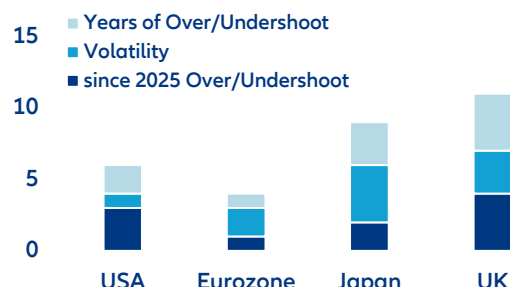
In years, last 20 years



Notes: Market-implied: 5y5y inflation swap, 10y inflation swap, 10 breakeven inflation, Surveys: NY Fed, Michigan University, Cleveland Fed, BoE, ECB, BoJ. Overshoot: long-term expectations > 2,15%; Undershoot: long-term expectations < 1,75%

Figure 11: Composite Score: Market- and survey- based Credibility Risk

Low = Low credibility risk; High = High credibility risk



Sources: LSEG Datastream, Allianz Research

Taken alone, neither market prices nor surveys measure the expectations that bear on central-bank credibility. Market-implied measures are prices, not beliefs: risk and liquidity premia, hedging demand and balance-sheet constraints can displace them by tens of basis points without any revision of expectations. The distortion is largest especially under stress, when the reading matters most.² They are also subject to Goodhart's law – once agents learn that a central bank reacts to a particular price, that price begins to reflect the anticipated reaction function, rather than belief.³ Surveys extend coverage to households and price-setting firms, but at the cost of a very short effective memory, which leaves them unable to separate a transitory relative-price shock from an erosion of the nominal anchor.⁴ We therefore complete them in two steps. First, we redirect market data from the level of expected inflation to regime and intermediation signals:

² Distortions arise from inflation risk and liquidity premia, regulation-driven hedging of real liabilities, central bank balance-sheet operations and for OTC inflation swaps counterparty credit and the funding cost of collateral. One illustration: the euro 5y5y fell to roughly 0.75% in March 2020 at the height of the dealer-intermediation dislocation. See D'Amico, Kim and Wei (2018); Fleckenstein, Longstaff and Lustig (2014)

³ The euro 5y5y is the clearest case. Mario Draghi elevated it to a quasi-official credibility gauge at Jackson Hole in August 2014; after its failure in 2020, the ECB's 2021 strategy review demoted it, admitting market-based measures only after model-based adjustment for risk premia and alongside the SPF, Consensus Economics and the Consumer Expectations Survey. See Work stream on inflation expectations, ECB Occasional Paper No. 264 (2021).

⁴ Household expectations form from salient recent prices, especially food and energy, even when the question specifies a long horizon; US households have additionally displayed partisan divergence as inflation has become a political subject. Professional forecasters show the opposite status quo bias, updating too slowly. See D'Acunto, Malmendier, Ospina and Weber (2021); Mankiw, Reis and Wolfers (2003)

our *Credibility Pressure*. Second, we model expectation formation as an adaptive process with an error-dependent estimation window: our *Expectation Credibility*.⁵

Central-bank credibility pressure has normalized, but pockets of instability remain

Instead of naively deriving inflation expectations from market prices, we define price stability as the central bank capacity to preserve the value of money under stress. We define price stability as the central bank's capacity to preserve the value of money, and measure the pressure on it across Mehrling's four prices of money.⁶ The aggregate is our *Credibility Pressure* indicator. It reads the regime a central bank operates in, rather than the level of any inflation price.

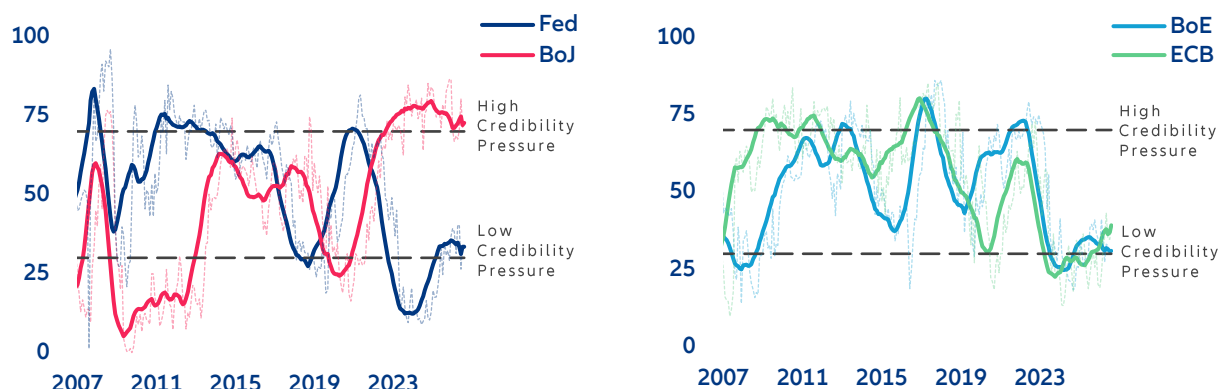
1. *The Par Value*: This is defined as the law of one price between private money and public money. We proxy it using the spread between the three-month Treasury bill yield, representing public money, and the three-month OIS rate, representing the risk-free price of bank-created money. Any widening of this spread signals tensions within the monetary system and growing doubts about the central bank's ability to keep short-term rates anchored. This measure has gained importance with the emergence of the *Activist Treasury Issuance* (ATI) in the US. Heavy bills issuance represents an expansion of public money supply from the Treasury, rather than the monetary authority. While the Fed's reverse repo facility (RRP) carried a large balance, that supply was absorbed without a price effect. With the facility drained, incremental bill issuance has to be taken down by real money, and the price appears in the Bill - OIS spread. The marginal price of the safest short rate is now influenced by the Treasury, not the central bank alone. This could lead to fiscal dominance visible at the front end. The *Par Value* would be the earliest to register it. Currently, the *Par Value* indicates no danger of front end fiscal dominance, but the US and UK are structurally more inclined to it (Figure 13).
2. *The Price of Goods*: We measure this as the spread between inflation rate and the overnight risk-free rate. This spread indicates whether inflation is running above or below the effective monetary stance. A widening positive gap suggests increasing pressure on the central bank's ability to maintain price stability and re-anchor inflation expectations.
3. *The Foreign Price of Money*: This is defined by the Real Effective Exchange Rate (REER), which captures the inflation-adjusted value of a currency against a basket of trading partners. It measures the external purchasing power and competitiveness of a currency. Sustained deviations from its historical equilibrium can signal growing doubts about the central bank's ability to preserve the external value of money.
4. *The Future Price of Money*. We proxy this through the steepening of the long end of the yield curve, measured by the 5y30y term spread. When the long end rises significantly relative to intermediate maturities, investors may be questioning the central bank's ability to control long-term rates and risk premia, thereby putting pressure on its credibility.

During the post-pandemic inflation shock, the Fed, the ECB and the BoE experienced significant credibility pressure. For all three, the pressure originated primarily from the *Price of Goods* (inflation) and *Future Price of Money* (curve steepness) dimensions of money. Following the aggressive monetary tightening since 2023, credibility pressure has eased and almost returned to neutral territory by now. Pockets of pressure remain in the steepening of the long end of the yield curve. (Figure 12).

⁵ Consistent with the experience-weighting evidence, see Malmendier and Nagel (2016) and adjacent to endogenous-gain learning, see Orphanides and Williams (2005)

⁶ Mehrling, P. (2023)

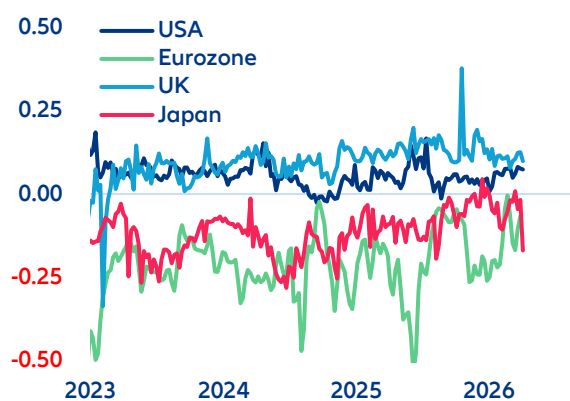
Figure 12: Evolution of the central-bank credibility pressure since 2007
Standardized composite Index



Sources: LSEG Datastream, Allianz Research

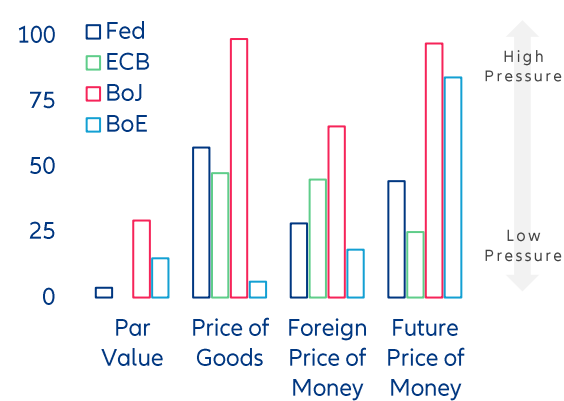
For the BoJ, the situation looks different. Having delayed the monetary response to inflation pressures, its credibility pressure has mounted across three dimensions simultaneously: JPY depreciation has weakened the foreign price of money, inflation overshoot affects the price of goods and the steepening of the long end of the yield curve suggests growing uncertainty regarding the future value of money. While tensions in the domestic monetary system remain relatively contained, the Bank of Japan's credibility is today the weakest in this framework. (Figure 14).

Figure 13: No fiscal dominance at short end for now, but US and UK with higher risk
Par Value: Bill-OIS spread, in pp



Sources: LSEG Datastream, Allianz Research

Figure 14: Current composition of central-bank credibility pressure
Standardized composite Index



More broadly, credibility pressure rarely emerges uniformly across all dimensions. Instead, it typically originates in one dimension before spreading to the others. By identifying the initial source of pressure and monitoring its transmission, *Credibility Pressure* provides a dynamic measure of effective central-bank credibility rather than monitoring level of inflation prices. It can also help to detect early signs of potential policy constraints before they become visible through conventional indicators

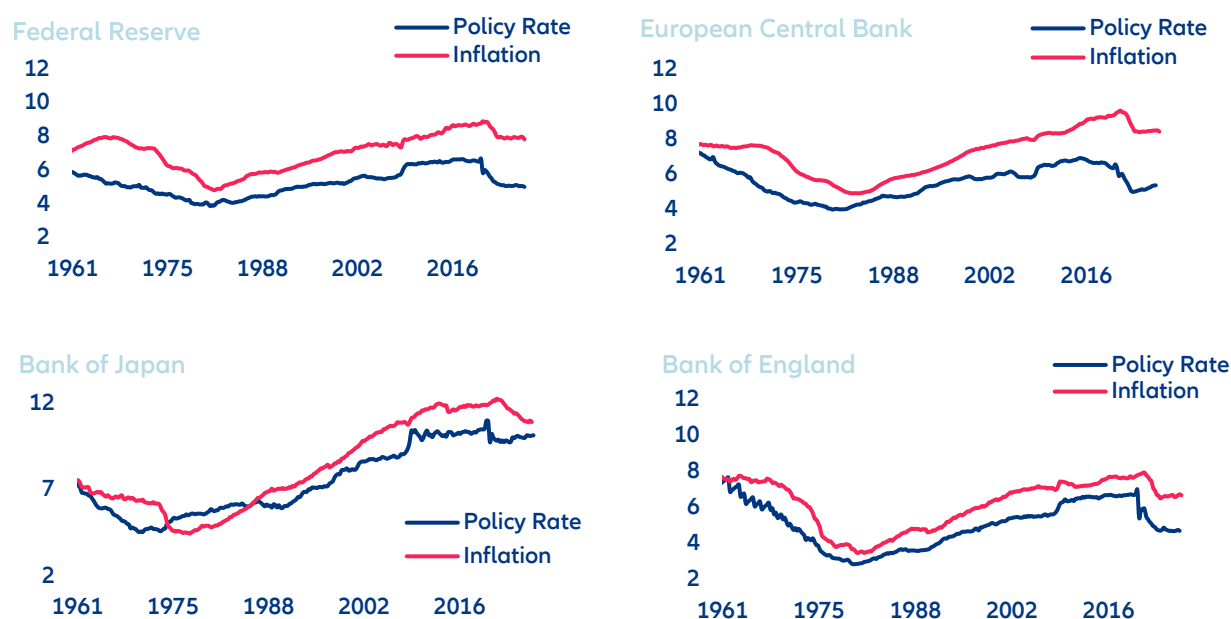
The broken of stability of monetary policy expectations

Monetary policy expectations have entered a fragile consolidation after the post-Covid inflation shock.

Since policy operates through expectations, credibility has to be measured through them instead of solely relying on noisy market-implied and survey-based indicators. We construct expectations as a weighted average of past observations in which the rate of forgetting rises with the magnitude of recent forecast errors: memory contracts under surprise and lengthens with accuracy.⁷ The estimation window that results is the credibility measure. It reads as how far back agents still find the past informative. An eight-year window means expectations rest on roughly the last eight years of data before memory decays materially, a four-year window means that everything before the last cycle has been discarded. Two structural features stand out. Inflation expectations are consistently formed over longer horizons than policy rate expectations: agents trust the inflation regime further out than they trust the reaction function. And credibility hit a global low in the early 1980s, when surging inflation and a break in the policy response cut the average window from eight years to four. Through the Great Moderation, central banks rebuilt it steadily, uninterrupted by the GFC. The peak came in 2021: eight years for inflation expectations in the UK and US, nine in the Eurozone, 12 in Japan; seven years for policy rates at the BoE, Fed and ECB, again 12 at the BoJ. The 2022 inflation shock broke that trend. Inflation memory shortened by up to two years, back to early-2000s levels, and the adjustment in policy rate expectations was sharper still, particularly in the UK and US. The BoJ was the exception. Expectation credibility is now consolidating, but precariously. Another inflation or deflation shock would take it below anything seen since the 1980s (Figure 15).

Figure 15: Credibility as memory: expectation windows for inflation and policy rates

Estimation window for adaptive filtered trend, in years



Sources: LSEG Datastream, Allianz Research

The credibility gap: Central banks have not yet regained their grip on the yield curve

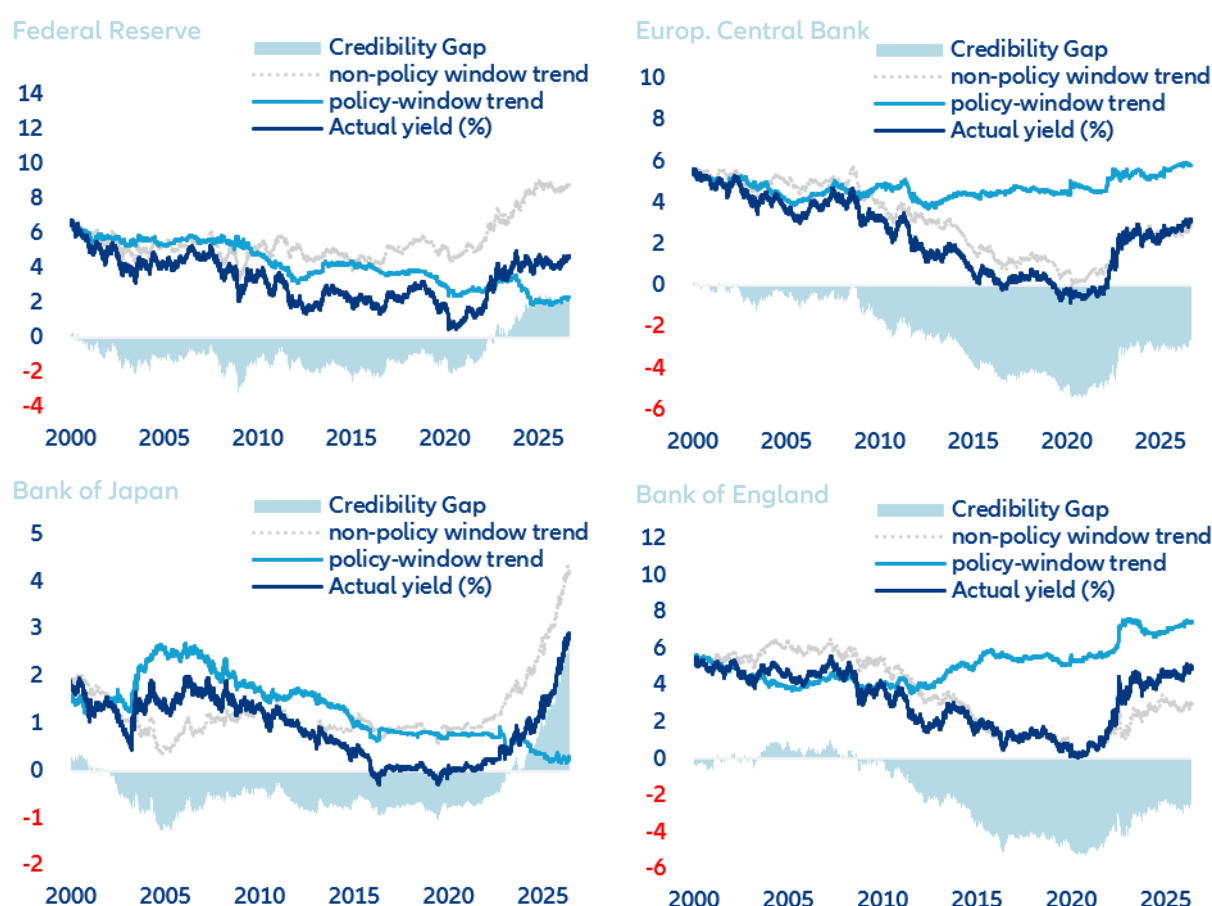
The balance-sheet grip has gone and rate decisions have not replaced it. We measure credibility by the extent to which a central bank is able to effectively set the level of yields. To measure this "grip on the yield curve" we use a decomposition that splits cumulative yield changes into those realized around policy meetings (t-2 to t+2) and those realized in between.⁸ To capture the long-term anchoring of yield levels, we use the 10-year maturity. The more the meeting window determines the trend in actual long-term yields, the

⁷ Barthalon, E., (2014)

⁸ Hillenbrand, S. (2025)

better the central bank is able to anchor rate expectations. The difference between the observed yield and the policy-window trend is thus a simple estimate of a "credibility gap" measuring how far the market has walked away from the central bank's anchor. For a credible central bank, the anchor holds between meetings and intermediate price moves are mostly offset. When the gap opens and stays open, policy meetings are unable to re-anchor the trend and the market sets long-term rates in response to other signals. For the US, between 2000 and 2008, the FOMC meeting window did a decent job of setting the long-term trend. When the Zero Lower Bound was reached and QE was introduced, the grip on the yield curve moved from rate decisions to the balance sheet. Since QE ended, the Fed's credibility gap has switched signs and widened progressively. The evolution is similar for the BoJ, where the surge came with the end of yield curve control in 2024 and is still deteriorating rapidly. The ECB and the BoE show a wide, negative gap that also opened with the ZLB and QE. The credibility gap analysis shows that central banks find themselves in a place where the balance-sheet grip has gone and has not yet been replaced by a grip through policy rate decisions (Figure 16).

Figure 16: QE is over, but policy rate decisions are yet unable to re-anchor long-term yield trends
10y yield, cumulative policy-window, non-policy window path and credibility gap, in %



Sources: LSEG Datastream, Allianz Research

The probability of a return to duration purchases, or of other measures to regain a grip on the curve, is higher in the US and Japan. One might object that this is not a loss of monetary grip, but a term premium rebuilding on fiscal supply and inflation risk. The timing argues otherwise. Term premium repricing was a common movement that began in 2022 and ran across all four jurisdictions more or less simultaneously. The credibility gaps did not. Each opened at an important change in that central bank's own balance sheet regime (start of QE for the ECB and the BoE, end of QE for the Fed and end of yield-curve control for the BoJ). A common shock does not produce staggered responses. The term premium is an enhancing factor, not the cause. Today all four face similarly sized gaps but with different signs, positive for the Fed and BoJ, negative

for the ECB and BoE. Closing them requires either regaining influence over the trend through a credible and legible reaction function, or acting on the yield itself through the balance sheet. The sign determines the urgency. Where the gap is positive, as for the Fed and the BoJ, the market is holding long yields above the policy-implied trend and closing it means buying duration. Where the gap is negative, as for the ECB and the BoE, the pressure runs the other way and is less acute. The probability of a return to duration purchases, or of other measures to regain grip on the curve, is therefore materially higher in the US and Japan. Markets may be underpricing this. A further credibility risk would be if Treasury were to massively buy back long-dated debt while issuing bills extracting duration and compressing the term premium as under a central-bank-orchestrated QE. In that case, the grip on the curve would be entirely assumed by Treasury setting the short-term bills issuance rate and steering long-term yields via buy-backs.

ECB with highest credibility, followed by the Fed. BoJ with highest credibility risk

The ECB scores highest, followed by the Fed. The credibility risk is structural for the BoE, but cyclical for the Fed and the BoJ. When we apply our six measures of central-bank credibility and establish an equally weighted ranking among the G4 central banks, the ECB scores equally well in structural and cyclical measures, emerging on top, while Fed credibility is strong structurally but weaker cyclically. The credibility of the ECB is an asset that could be used to strengthen the international role of the euro.⁹ In comparison, the BoE shows an elevated credibility risk, suffering notably from weak scores in structural measures such as expectations, anchoring and institutional framework. The BoJ shows the highest credibility risk, suffering from weak scores in cyclical indicators such as credibility sentiment and pressure. This means that any premium that is priced for central-bank credibility is more of a cyclical nature for the BoJ and the Fed, and is more structural for the BoE (Table 1).

Table 1: The Central Bank credibility framework

Ranking by subcategory and aggregate Ranking – the lower the score the lower the credibility risk

Measure	Type	Fed	ECB	BoJ	BoE
Insitutional framework	Structural	2	1	3	4
Credibility Risk Sentiment	Cyclical	3	2	4	1
Markets and Surveys	Cyclical	2	1	3	4
Credibility Pressure	Cyclical	2	3	4	1
Expectations Anchoring	Structural	3	2	1	4
Credibility Gap	Structural	3	1	4	2
Total			2,5	1,7	3,2

Source: Allianz Research

How to restore central bank credibility?

The withdrawal of forward guidance is not a credibility risk if communication is embedded in an “explain or comply” framework. Central banks hold no informational advantage as public data usually forecast policy rates and inflation about as well as markets do.¹⁰ Guidance is therefore always part forecast and part promise, and withdrawing forward guidance is not in itself a credibility loss. What a central bank does know better than the market is its own reaction function. Publishing a benchmark rule, explaining deviations from it and releasing vote tallies as a gauge of how stable the rule is and how long a deviation is expected to last would be the genuine credibility enhancer.

Develop a more nominal framing of policy targets. Under inflation targeting, a supply shock forces a choice between tolerating a strong inflation overshoot and crushing output; under nominal GDP (NGDP) level targeting, the framework answers for itself since the nominal target is left broadly unchanged. A more

⁹ Rey, H. and Subran, L. (2025), [Link](#)

¹⁰ Giannone, Reichlin and Sala (2004)

nominal framing, including nominal wages, also makes communication legible to the wider public. But the switch is itself the credibility cost. The better route is to publish nominal wage growth and NGDP as reference variables against which decisions are explained, leaving the mandate untouched.

Prepare for the next credibility challenge. The recent strategic reviews adapted to a world of supply shocks: the Fed abandoned average inflation targeting, the BoJ abandoned its inflation-overshooting commitment. Both frameworks were built for a persistent undershoot that failed its first supply shock. We see that as a sign that the prevailing economic culture in central banks, and the models that embed it, have shifted decisively towards inflation vigilance. Should deflationary pressure return (AI-driven supply expansion and productivity gains, demographics, Chinese excess capacity), the available instruments may prove neither adequate nor prompt. Strategic reviews on a fixed schedule tend to be procyclical and cap the horizon agents are willing to believe in. The lower-bound answer is best decided now, while it costs nothing, rather than at the bound.

Manage the relationship with the Treasury. This is mainly a US question today, but potentially a broader one as debt-service costs rise. If issuance composition can offset monetary policy, the policy rate stops summarizing the stance and forecasting monetary policy becomes a two-player problem. For the Treasury, the remedy is a public recommitment to regular and predictable issuance. For the central bank, it is a stance measure that internalizes Treasury duration supply. Making ATI visible makes it less challenging for central bank credibility.

These assessments are, as always, subject to the disclaimer provided below.

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